ENVIRONMENTAL SCIENCE: THEORY INTO PRACTICE-II

SYLLABUS:

UNIT-5
Global environmental issues and policies

UNIT-6
Biodiversity and Conservation4

UNIT-7
Human communities and the Environment

For regular students

- ⇨ Part A- 30 Marks

For sol students

- ⇨ Part A- 30 Marks
- ⇨ Part B- 20 Marks
- ⇨ Part C- 20 Marks

QUES 1 IS COMPULSORY

UNIT-V

GLOBAL ENVIRONMENTAL ISSUES AND POLICIES

- Causes of climate change, Global Warming, Ozone layer depletion and acid rain; Impacts on human communities, biodiversity, global economy and agriculture.
- International agreements and programmes: Earth summit, UNFCCC, Montreal and Kyoto protocols, convention on biological diversity, Ramsar convention, The Chemical weapons conventions, UNEP, CITIES etc
- Sustainable development goals: India's National Action Plan on climate change and its major missions
- Environment legislation in India: Wildlife Protection Act, 1972; Water (Prevention and control of pollution) Act, 1974; Forest (Conservation) Act 1980; Air (prevention and control of pollution) Act,1981; Environment protection Act, 1986; Scheduled tribes and other traditional forest dwellers (Recognition of forest rights) Act, 2006.

UNIT - VI

BIODIVERSITY AND CONSERVATION

- Definition of biodiversity; Levels of biological diversity: Genetic, species and ecosystem diversity.
- India as a mega-biodiversity nation; Biogeographic zones of India; Biodiversity hotspots; endemic and endangered species of India; IUCN Red list criteria and categories.
- Value of biodiversity; Ecological, economic, social, ethical, aesthetic and informational values of biodiversity with examples; sacred groves and their importance with examples.
- Threats to biodiversity: Habitat loss, degradation and fragmentation; Poaching of wildlife; Man-wildlife conflicts; Biological invasion with emphasis on Indian biodiversity; Current mass extinction crisis.
- Biodiversity conservation strategies: In-situ and Ex-situ methods of conservation; National parks, Wildlife sanctuaries, and Biosphere reserves; keystone, flagship, umbrella, and indicator species; Species reintroduction and translocation.
- Case studies: Contemporary Indian wildlife and biodiversity issues, movements and projects (e.g., Project Tiger, Project Elephant, Vulture Breeding program, Project Great Indian bustard, Crocodile conservation project, Silent valley movement, Save western ghats movement, etc)

UNIT – VII

HUMAN COMMUNITIEES AND THE ENVIRONMENT

- Human population growth: Impacts on environment, human health, and welfare; Carbon foot-print.
- Resettlement and rehabilitation of development project affected persons and communities; relevant case studies.
- Environmental movements: Chipko movement, Appiko Movement, Silent valley movement, Bishnois of rajasthan, Narmada bachao Andolan, etc.
- Environmental justice: National Green tribunal and its importance.
- Environmental philosophy: Environmental ethics; role of various religions and cultural practices in environmental conservation.
- Environmental communication and public awareness: Case studies (e.g., CNG vehicles in Delhi, Swachh Bharat Abhiyan, National Environment Awareness campaign (NEAC) , National green corps (NGC) " ECO-CLUB" programme, etc.)

UNIT-V

GLOBAL ENVIRONMENTAL ISSUES AND POLICIES

Q1. What is climate change and list the causes of climate change

Climate change refers to the **long-term change in the average weather conditions** of the Earth, such as **temperature, rainfall, wind patterns and seasons**, over a long period of time. It is mainly caused due to **human activities** that increase the concentration of **greenhouse gases** in the atmosphere, leading to **global warming**.

Causes of Climate Change

The major causes of climate change are as follows:

1. **Burning of Fossil Fuels**
   Excessive use of coal, petroleum and natural gas in **industries, power plants and vehicles** releases large amounts of **carbon dioxide (CO_2)**, a major greenhouse gas.
2. **Deforestation**
   Cutting of forests reduces the number of trees that absorb CO_2 from the atmosphere, resulting in **increased greenhouse gases**.
3. **Industrialization**
   Industries release harmful gases such as **carbon dioxide, methane and nitrous oxide**, which contribute to global warming.
4. **Agricultural Activities**
   Use of chemical fertilizers and paddy cultivation release **methane and nitrous oxide**, increasing atmospheric pollution.
5. **Urbanization and Population Growth**
   Rapid growth of cities increases energy consumption, waste generation and vehicular emissions, accelerating climate change.
6. **Use of CFCs and Other Chemicals**
   Refrigerators, air conditioners and aerosol sprays release **chlorofluorocarbons (CFCs)**, which trap heat and damage the atmosphere.

Q2. What is Global Warming? And what are its effects?

Global warming refers to the **gradual increase in the average temperature of the Earth's atmosphere** due to excessive accumulation of **greenhouse gases**. These gases trap heat from the sun and prevent it from escaping into space, resulting in the **greenhouse effect**.

Contribution of Different Gases to Global Warming

Different greenhouse gases contribute to global warming in varying proportions:

1. **Carbon Dioxide (CO_2) – about 60%**
   - Released from burning fossil fuels, vehicles, industries and deforestation
   - Major contributor to global warming
2. **Methane (CH_4) – about 15–20%**
   - Emitted from agriculture, paddy fields, landfills and livestock
   - More heat-trapping than CO_2
3. **Chlorofluorocarbons (CFCs) – about 12–15%**
   - Released from air conditioners, refrigerators and aerosol sprays

- Strong greenhouse gases with long atmospheric life

4. **Nitrous Oxide (N_2O) – about 6%**
   - Released from fertilizers, industries and combustion processes
   - Has high global warming potential
5. **Water Vapour – variable contribution**
   - Naturally occurring gas
   - Increases the warming effect once temperature rises

Effects of Global Warming

1. **Rise in Global Temperature**
   Leads to hotter summers and frequent heat waves.
2. **Melting of Glaciers and Polar Ice Caps**
   Causes **rise in sea level**, threatening coastal areas.
3. **Climate Change and Extreme Weather Events**
   Increased floods, droughts, cyclones and irregular rainfall.
4. **Impact on Agriculture**
   Reduced crop productivity due to unpredictable weather and water scarcity.
5. **Loss of Biodiversity**
   Many plant and animal species face extinction due to habitat loss.
6. **Effects on Human Health**
   Increase in heat-related diseases, respiratory problems and spread of infections.

Effects of Global Warming on Agriculture

1. **Irregular Rainfall**
   Global warming disturbs rainfall patterns, causing floods and droughts, which negatively affect crop production.
2. **Reduction in Crop Yield**
   Increased temperature lowers the productivity of major crops like wheat and rice.
3. **Increased Pest and Disease Attacks**
   Warmer climate promotes the growth of pests and plant diseases, damaging crops.
4. **Water Scarcity for Irrigation**
   Changing climate and melting glaciers reduce water availability for agricultural use.

Q3. Ozone layer depletion and its impact

Ozone layer depletion refers to the **thinning of the ozone layer** present in the stratosphere due to the release of harmful chemicals like **chlorofluorocarbons (CFCs)**. The ozone layer protects the Earth by **blocking harmful ultraviolet (UV) rays** of the sun.

Impacts of Ozone Layer Depletion

1. **Human Health Problems**
   Increased UV radiation causes skin cancer, eye cataracts and weakens the immune system.
2. **Damage to Crops and Plants**
   UV rays reduce photosynthesis, leading to lower crop productivity.
3. **Harm to Marine Ecosystem**
   Phytoplankton are affected, disturbing the aquatic food chain.

4. **Environmental Imbalance**
   Disturbs climate and ecosystems, leading to biodiversity loss.

Q4. Acid rain and its impact

Acid rain refers to **rainfall that contains harmful acids**, formed when **sulphur dioxide (SO_2)** and **nitrogen oxides (NO_x)** released from industries, vehicles and power plants mix with water vapour in the atmosphere.

Causes of Acid Rain

- Burning of coal and petroleum in industries
- Emissions from automobiles
- Thermal power plants and factories

Impacts of Acid Rain

1. **Damage to Crops and Forests**
   Acid rain damages leaves, reduces soil fertility and lowers agricultural productivity.
2. **Harm to Aquatic Life**
   Acidic water kills fish and other aquatic organisms in lakes and rivers.
3. **Soil Degradation**
   Leaching of essential nutrients like calcium and magnesium from soil.
4. **Damage to Buildings and Monuments**
   Corrodes marble and limestone structures (e.g., Taj Mahal).

Q5. Explain the natural and anthropogenic causes of climate change.

Natural and Anthropogenic Causes of Climate Change

Climate change refers to long-term changes in temperature, precipitation and other climatic patterns. It is caused by **natural processes** as well as **human activities**.

A. Natural Causes of Climate Change

1. Solar Variability

- The Sun is the primary source of energy for Earth.
- Variations in solar radiation influence Earth's climate over long periods.

(a) Milankovitch Cycles

- Long-term changes in Earth's:
  - Orbital shape (eccentricity)
  - Axial tilt
  - Axial precession (wobble)
- These affect the amount and distribution of solar energy received by Earth.

(b) Solar Cycles

- The Sun has an approximately **11-year sunspot cycle**.
- Changes in solar activity can influence atmospheric circulation and regional climates.

2. Volcanic Activity

- Volcanic eruptions release **ash, dust and sulphur dioxide** into the atmosphere.
- These particles block sunlight, causing **temporary cooling**.
- Example: **Mount Tambora eruption (1815)** led to the "Year Without a Summer".

3. Ocean Circulation Patterns

- Changes in ocean currents affect global climate.

- **El Niño–Southern Oscillation (ENSO)** alters temperature and rainfall patterns worldwide.
- Influences droughts, floods and extreme weather events.

4. Geological Processes

- Plate tectonics and mountain formation influence climate.
- Mountain ranges affect wind flow and rainfall.
- Example: **Himalayas influence the Asian monsoon system**.

5. Natural Greenhouse Gas Emissions

- Greenhouse gases are released naturally and are part of the Earth's carbon cycle.

(a) Methane

- Released from wetlands and thawing permafrost.
- Methane is a powerful greenhouse gas.

(b) Carbon Dioxide

- Released during volcanic eruptions and natural forest fires.

B. Anthropogenic (Human-Induced) Causes of Climate Change

Human activities are the **main cause of present-day global warming**.

1. Burning of Fossil Fuels

- Combustion of coal, oil and natural gas releases large amounts of **carbon dioxide (CO_2)**.

Major sources:

- Electricity generation
- Transportation (cars, trucks, aircraft)
- Industrial activities
- Residential and commercial heating

2. Deforestation and Land-Use Change

- Forest clearing for agriculture, urbanization and infrastructure.

Effects:

- Release of stored CO_2
- Loss of carbon sinks
- Alteration of local and regional climate patterns

3. Agriculture and Livestock Farming

- Significant source of greenhouse gases.

Main emissions:

- **Methane** from cattle digestion and manure
- **Nitrous oxide** from chemical fertilizers
- CO_2 from land conversion and farm machinery

4. Industrial Processes

- Energy-intensive industries like cement, steel and chemicals emit GHGs.
- Certain chemical reactions release methane and other gases.

5. Waste Generation

- Poor waste management contributes to climate change.

Sources:

- Methane from landfills
- Methane from wastewater treatment plants

6. Use of Refrigerants

- Air conditioners and refrigerators use **HFCs and HCFCs**.
- These gases have very high global warming potential.

7. Land Degradation

- Caused by overgrazing, deforestation and poor farming practices.
- Leads to soil erosion, desertification and release of stored carbon.

8. Urbanization

- Expansion of cities increases energy consumption.
- Causes the **Urban Heat Island effect** due to concrete and asphalt.

9. Transportation

- Vehicles using fossil fuels emit CO_2.
- Infrastructure development also increases emissions.

10. Consumption Patterns

- High consumption of energy-intensive goods increases carbon footprint.
- Lifestyle choices affect demand for fossil fuels and natural resources.
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Q6.

Environmental Issues and Farmer Suicides in India

Environmental problems, especially climate change, have significantly contributed to the rising number of farmer suicides in India. Agriculture in India is highly dependent on natural factors such as rainfall, temperature and soil quality, making farmers extremely vulnerable to environmental stress.

A. Environmental Issues Responsible for Increasing Farmer Suicides

1. Climate Change and Rising Temperatures

- Climate change has increased average temperatures across India.
- Studies show that a rise of **1°C during the crop-growing season** leads to a significant increase in farmer suicides.
- High temperatures damage crops, reduce productivity and increase crop failure.

2. Erratic Rainfall and Droughts

- Monsoon rainfall has become irregular and unpredictable.
- Frequent droughts lead to crop failure and loss of income.
- Rain-fed agriculture dominates Indian farming, especially among small and marginal farmers.

3. Floods and Extreme Weather Events

- Sudden floods destroy standing crops, stored grains and agricultural infrastructure.
- Cyclones and unseasonal rains increase financial losses.
- Farmers often lack insurance coverage for such disasters.

4. Declining Soil Fertility

- Soil degradation due to overuse of chemical fertilizers and pesticides.
- Loss of soil nutrients reduces crop yield and profitability.
- Increased cost of fertilizers pushes farmers into debt.

5. Water Scarcity

- Falling groundwater levels due to excessive extraction.
- Inadequate irrigation facilities worsen the situation.
- Crop stress due to lack of water leads to economic distress.

6. Pest Attacks and Crop Diseases

- Rising temperatures and humidity increase pest infestations.
- Farmers spend more on pesticides, increasing input costs.
- Failure to control pests leads to crop losses.

7. Economic Stress Due to Environmental Losses

- Repeated crop failures result in mounting debts.
- Inability to repay loans leads to mental stress and depression.
- Environmental uncertainty increases insecurity among farmers.

B. Measures the Government Should Take to Improve the Agricultural Sector

1. Development of Irrigation Facilities
- Expand canal irrigation and drip irrigation systems.
- Promote rainwater harvesting and watershed management.
- Reduce dependence on monsoon rainfall.

2. Climate-Resilient Agriculture
- Promote drought-resistant and heat-tolerant crop varieties.
- Encourage crop diversification to reduce risk.
- Support climate-smart farming techniques.

3. Fair Pricing and Income Support
- Ensure **Minimum Support Price (MSP)** for all major crops.
- Timely procurement of crops by the government.
- Introduce direct income support schemes for farmers.

4. Affordable Credit and Debt Relief
- Provide easy and timely loans at low interest rates.
- Strengthen cooperative banks and rural credit institutions.
- Implement effective debt waiver schemes during disasters.

5. Crop Insurance and Risk Coverage
- Improve crop insurance schemes with faster claim settlement.
- Cover losses due to climate-related disasters.
- Increase awareness about insurance among farmers.

6. Promotion of Sustainable Farming Practices
- Encourage organic farming and reduced chemical usage.
- Promote soil conservation and integrated pest management.
- Improve long-term soil health and productivity.

7. Farmer Education and Awareness
- Conduct training programs on modern agricultural practices.
- Set up demonstration farms to showcase new technologies.
- Use digital platforms to provide weather and market information.

8. Alternative Livelihood Opportunities
- Encourage mixed farming (crop + livestock).
- Promote agro-based cottage industries.
- Reduce total dependence on agriculture for income.

Q7.

Importance of Wetlands with regard to Biodiversity and Water Conservation

Meaning of Wetlands
- Wetlands are land areas that are **permanently or seasonally saturated with water**.
- They include **ponds, lakes, rivers, estuaries, mangroves, marshes, swamps, reservoirs and creeks**.
- Types of wetlands:

- **Marine wetlands** – coastal areas
- **Estuarine wetlands** – deltas, tidal marshes, mangroves
- **Lacustrine wetlands** – lakes
- **Riverine wetlands** – rivers and streams
- **Palustrine wetlands** – marshes, swamps, bogs

A. Importance of Wetlands for Biodiversity

- Wetlands are **biodiversity hotspots** due to high productivity and nutrient recycling.
- They support **aquatic plants, fishes, amphibians, reptiles, birds and mammals**.
- Freshwater ecosystems in India support large biological diversity:
    - Western Ghats freshwater systems support about **290 species of fish**.
- Wetlands act as **breeding and nursery grounds** for many species.
- They are critical habitats for **migratory birds**:
    - Bharatpur Bird Sanctuary (Rajasthan)
    - Little Rann of Kutch and coastal Gujarat
- Provide shelter to **endangered species**:
    - **Sangai deer** in Loktak Lake (Manipur)
    - **Sarus crane**, **Black-necked crane**
    - **Gangetic river dolphin**
    - **Indian mud turtle**
- Wetlands maintain **food chains and ecological balance**.

B. Importance of Wetlands for Water Conservation

1. Water Purification

- Wetlands act as **natural filters**.
- They trap sediments and remove pollutants, toxins and excess nutrients.
- Reduce water treatment costs.

2. Flood Control

- Wetlands act like **natural sponges**.
- They absorb excess rainwater and reduce flooding.
- Slow release of water prevents sudden floods downstream.

3. Groundwater Recharge

- Wetlands allow water to **percolate into the soil**.
- Recharge underground aquifers.
- Maintain water availability during dry periods.

4. Water Storage

- Wetlands store water during rainy seasons.
- Release water gradually during droughts.
- Ensure continuous water supply for ecosystems and humans.

5. Climate Regulation

- Wetlands help in **carbon sequestration**.
- Reduce impacts of climate change.
- Maintain local temperature and humidity.

Q8.

Short Note on Ramsar Convention

- The **Ramsar Convention** is an **international treaty for wetland conservation**.
- It was adopted in **1971 at Ramsar, Iran**.

- Objective:
    - Conservation and **wise use of wetlands** through national action and international cooperation.
- India is a **signatory member** of the convention.
- India has **26 designated Ramsar Sites**, with more under consideration.

Definition of Wetlands under Ramsar Convention

- Includes:
    - Lakes and rivers
    - Marshes and swamps
    - Peatlands and mangroves
    - Estuaries and tidal flats
    - Coral reefs
    - Human-made wetlands like rice fields, reservoirs, fish ponds and salt pans

Three Pillars of Ramsar Convention

1. **Wise use of all wetlands**
    - Sustainable utilization without degradation
2. **Designation of Ramsar Sites**
    - Identify wetlands of international importance
    - Ensure their effective management
3. **International cooperation**
    - Protect shared wetlands, migratory species and transboundary water systems

Q9. Comment on the following

(a) Earth Summit (Rio Earth Summit, 1992)

- Official name: **United Nations Conference on Environment and Development (UNCED)**
- Held at **Rio de Janeiro, Brazil**, from **3–14 June 1992**
- Aim:
    - To balance **economic development** with **environmental protection**
    - To promote **sustainable development**
- Attended by:
    - **117 Heads of State**
    - Representatives from **178 countries**
- Major outcomes:
    - **Rio Declaration on Environment and Development**
    - **Agenda 21** (global action plan for sustainability)
    - **Forest Principles**
- Major conventions opened for signature:
    - **UNFCCC**
    - **Convention on Biological Diversity (CBD)**
    - **UN Convention to Combat Desertification**
- Led to later agreements like:
    - **Kyoto Protocol**
    - **Paris Agreement**
- Criticism:

- Many commitments were not fully implemented, especially on poverty and environment

(b) UNFCCC (United Nations Framework Convention on Climate Change)

- Adopted on **9 May 1992**
- Opened for signature at **Earth Summit, Rio**
- Entered into force on **21 March 1994**
- Objective:
  - To **stabilize greenhouse gas concentrations**
  - To prevent dangerous human interference with the climate system
- Nature:
  - Framework convention (non-binding emission targets)
- Members:
  - **197 Parties**
- Important developments under UNFCCC:
  - **Kyoto Protocol (1997)** – legally binding emission reduction targets
  - **Paris Agreement (2015)** – limit global warming to **1.5°C**
- Annual meetings:
  - **Conference of Parties (COP)**
- Secretariat headquarters:
  - **Bonn, Germany**

(c) CBD (Convention on Biological Diversity)

- Opened for signature at **Earth Summit, 1992**
- Entered into force on **29 December 1993**
- Objectives:
  - Conservation of **biological diversity**
  - **Sustainable use** of biological resources
  - **Fair and equitable sharing** of benefits from genetic resources
- Governing body:
  - **Conference of Parties (COP)** (meets every two years)
- Supplementary protocols:
  - **Cartagena Protocol** – Biosafety of Living Modified Organisms
  - **Nagoya Protocol** – Access and Benefit Sharing (ABS)
- Importance:
  - Encouraged global awareness about biodiversity
  - Key instrument for sustainable development
- Members:
  - **194 countries**

(d) Ramsar Convention

- Full name:
  - **Convention on Wetlands of International Importance**
- Signed on:
  - **2 February 1971** at **Ramsar, Iran**
- Objective:
  - Conservation and **wise use of wetlands**

- Covers:
    - Lakes, rivers, marshes, mangroves, estuaries
    - Human-made wetlands like reservoirs and rice fields
- Governing bodies:
    - **Conference of Contracting Parties (COP)**
    - **Standing Committee**
    - **Scientific and Technical Review Panel (STRP)**
    - **Secretariat** (based at IUCN, Switzerland)
- Importance:
    - Protection of wetlands and waterfowl habitats
- Celebrated as:
    - **World Wetlands Day – 2 February**

(e) CWC (Chemical Weapons Convention)

- Full name:
    - **Convention on the Prohibition of Chemical Weapons**
- Entered into force:
    - **29 April 1997**
- Objective:
    - Ban on **development, production, stockpiling and use** of chemical weapons
- Administered by:
    - **Organisation for the Prohibition of Chemical Weapons (OPCW)**
    - Headquarters: **The Hague, Netherlands**
- Key provisions:
    - Destruction of chemical weapons and production facilities
    - International inspections
    - Assistance to victims of chemical attacks
- Membership:
    - **193 States Parties**
- Achievement:
    - Over **96% of declared chemical weapons destroyed**

(f) UNEP (United Nations Environment Programme)

- Established in:
    - **1972**, after the **Stockholm Conference**
- Headquarters:
    - **Nairobi, Kenya**
- Role:
    - Coordinates environmental activities of the United Nations
- Objectives:
    - Promote environmental protection
    - Support sustainable development
- Major areas of work:
    - Climate change
    - Ecosystem management
    - Environmental governance
    - Disaster risk reduction

- Publications:
    - **Global Environment Outlook (GEO)**
- Contribution:
    - Development of international environmental treaties
    - Environmental data and policy support
- Also known as:
    - **UN Environment**

(G) CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora)

- Full name:
    - **Convention on International Trade in Endangered Species of Wild Fauna and Flora**
- Also known as:
    - **Washington Convention**
- Nature:
    - A **multilateral international treaty**
- Drafted:
    - In **1963** after a resolution by **IUCN (International Union for Conservation of Nature)**
- Opened for signature:
    - **1973**
- Entered into force:
    - **1 July 1975**
- Main objective:
    - To ensure that **international trade in wild animals and plants does not threaten their survival**
- Scope:
    - Regulates trade in:
        - Live animals and plants
        - Animal skins, leather goods
        - Food products, medicines and ornaments
- Protection:
    - Provides protection to **more than 35,000 species** of animals and plants
- Administration:
    - Secretariat administered by **UNEP**
    - Headquarters: **Geneva, Switzerland**
- Decision-making body:
    - **Conference of the Parties (CoP)**
- Membership:
    - **183 Parties** (as of June 2019)
- Legal status:
    - Legally binding but implemented through **national laws**
- Key features:
    - Licensing system for import, export and re-export
    - Management Authorities and Scientific Authorities appointed by each country
    - Species listed under **three Appendices** based on threat level

- Conferences:
    - CoP held every **2–3 years**
    - **17th CoP**: Johannesburg, 2016
    - **India hosted 3rd CoP in 1981**

(H) Rio Earth Summit, 1992

- Official name:
    - **United Nations Conference on Environment and Development (UNCED)**
- Venue:
    - **Rio de Janeiro, Brazil**
- Year:
    - **1992**
- Purpose:
    - To address the relationship between:
        - Environment
        - Economic development
        - Science and politics
- Participation:
    - Leaders from **105 nations**
- Core objective:
    - Promotion of **sustainable development**
- Major outcomes:
    - **Rio Declaration on Environment and Development**
    - **Agenda 21**
    - **Forest Principles**
- Importance:
    - Strengthened global environmental governance
    - Encouraged international cooperation
- Additional outcome:
    - Formation of **Green Cross International**

(I) Montreal Protocol

- Full name:
    - **Montreal Protocol on Substances that Deplete the Ozone Layer**
- Adopted:
    - **16 September 1987**
- Entered into force:
    - **1 January 1989**
- Objective:
    - To reduce and eliminate **ozone-depleting substances (ODS)**
- Major ODS covered:
    - Chlorofluorocarbons (CFCs)
    - Halons
    - Carbon tetrachloride
    - Methyl chloroform
- Goal:
    - Phase-out of ODS by **2000** (2005 for some substances)

- Importance:
    - Protects the **ozone layer** from harmful **UV-B radiation**
- Achievement:
    - Ozone hole over Antarctica is **slowly recovering**
- Future projection:
    - Ozone layer expected to return to **1980 levels by 2050–2070**
- Significance:
    - Considered one of the **most successful international environmental agreements**

(J) Kyoto Protocol

- Linked to:
    - **United Nations Framework Convention on Climate Change (UNFCCC)**
- Adopted:
    - **1997** (Kyoto, Japan)
- Entered into force:
    - **2005**
- Objective:
    - To reduce **greenhouse gas (GHG) emissions**
- Target group:
    - **Developed countries only**
- Emission reduction target:
    - Average **5% reduction** below 1990 levels
- Commitment period:
    - **2008–2012**
- Greenhouse gases covered:
    - Carbon dioxide (CO_2)
    - Methane (CH_4)
    - Nitrous oxide (N_2O)
    - HFCs, PFCs
    - Sulphur hexafluoride (SF_6)
- Principle followed:
    - **Common but Differentiated Responsibility**
- Doha Amendment:
    - Adopted on **8 December 2012**
    - Second commitment period: **2013–2020**
- Importance:
    - First legally binding treaty to combat climate change

Q10.

Sustainable Development

Meaning / Definition

- Sustainable development refers to **development that meets the needs of the present without compromising the ability of future generations to meet their own needs**.

- It involves **positive economic and social change** without disturbing the **ecological and environmental balance** of the biosphere.
- It ensures that development activities **do not degrade natural resources**, biodiversity, or life-support systems.

Origin of the Concept

- The concept was first highlighted at the **Stockholm Conference on Human Environment, 1972 (Sweden)**.
- The conference emphasized:
  - Global cooperation
  - Protection and enhancement of the human environment
  - Responsible use of natural resources

Relationship between Sustainability and Development

- **Sustainability** ensures long-term environmental balance.
- **Development** focuses on economic growth and improvement in quality of life.
- **Sustainable development integrates both**, ensuring growth without environmental degradation.

Broad Requirements of Sustainable Development

- **Integration of conservation and development**
- **Fulfilment of basic human needs** (food, shelter, water, health)
- **Social justice and equity**
- **Cultural diversity and self-determination**
- **Maintenance of ecological integrity**

Essential Conditions for Achieving Sustainable Development

- Human demands must remain **within the carrying capacity of the environment**.
- Conservation of resources for **future generations and other species**.
- **Reduction of poverty**.
- **Equitable distribution of resources and opportunities**.
- Participation of:
  - Indigenous communities
  - Women
  - Tribals
- **Efficient waste management and pollution control**.
- Access to **clean and renewable energy**.
- Protection of **marine life** from pollution and climate change.
- **Disaster resilience and preparedness**.
- **Controlled use of hazardous materials**.
- Investment in **health and education**.

International Efforts Linking Sustainable Development with Environment

1. Convention on Biological Diversity (1994)

- Objective:
  - Conservation of biodiversity
  - Sustainable use of biological resources
- Measures:
  - National conservation strategies
  - Public awareness programmes
  - Protection of endangered species

2. United Nations Framework Convention on Climate Change (1992)

- Signed at the **Rio Earth Summit**
- Aim:
    - Address global warming and climate change
- Measures:
    - Reporting of greenhouse gas emissions
    - Protection of forests and carbon sinks
    - Climate-resilient planning for agriculture, water and coastal zones

3. Principles of Forest Management
- Adopted at the **Rio Earth Summit**
- Emphasized:
    - Conservation of forests
    - Cultural, ecological and spiritual importance of forests
    - Sustainable forest management

4. Rio Declaration on Environment and Development (1992)
- Established the link between **economic development and environmental protection**
- Recognized:
    - Right to a healthy environment
    - Principle of **"Polluter Pays"**

5. Agenda 21
- A comprehensive **300-page action plan** for sustainable development
- Adopted by **100 heads of states**
- Focus areas:
    - Conservation of land, water and air
    - Poverty alleviation
    - Health and education
    - Sustainable consumption and production

6. World Summit on Sustainable Development (2002)
- Held at **Johannesburg**
- Strengthened and expanded sustainable development principles adopted at Rio (1992)

7. United Nations Climate Change Conference (2005)
- Held at **Montreal, Canada**
- Developed:
    - A five-year programme on climate change impacts
    - Encouraged international cooperation on climate action

Q11. What is Sustainability? Describe its three pillars. What are the indicators of sustainability?

Meaning of Sustainability
- The word **"sustain"** means *to maintain or endure*.
- **Sustainability** refers to the ability of human society to **survive and progress on Earth without degrading natural resources**.
- It ensures that **human needs and environmental limits are balanced**.
- According to a widely accepted definition:
    - *"Sustainability creates and maintains conditions under which humans and nature can exist in productive harmony, fulfilling the social, economic and environmental needs of present and future generations."*

- Everything required for human survival and well-being depends **directly or indirectly on the natural environment**.

Principles of Sustainability (Derived from Nature)

Nature sustains itself through four inter-connected principles:

1. **Solar Energy**
    - Sun is the primary source of energy for life.
    - Promotes use of renewable and clean energy.
2. **Population Control**
    - Population growth should remain within the carrying capacity of Earth.
    - Overpopulation leads to resource depletion.
3. **Nutrient Recycling**
    - Continuous recycling of nutrients (carbon, nitrogen, water cycles).
    - Reduces waste and conserves resources.
4. **Biodiversity**
    - Diversity of species increases ecosystem stability and resilience.

Three Pillars of Sustainability

The concept of sustainability rests on **three main pillars**:

1. Economic Sustainability

- Efficient and responsible use of resources.
- Ensures long-term economic growth without harming the environment.
- Aims at:
    - Employment generation
    - Income stability
    - Reduction of poverty
- Economic activities should be viable in the long run.

2. Social Sustainability

- Ability of society to maintain **social well-being and harmony**.
- Focuses on:
    - Equality and social justice
    - Access to education and healthcare
    - Participation of women, tribals and indigenous communities
- Ensures quality of life for present and future generations.

3. Environmental Sustainability

- Living within the limits of natural ecosystems.
- Ensures that natural resources are **used at a rate slower than their regeneration**.
- Emphasizes:
    - Conservation of air, water, soil and biodiversity
    - Pollution control
    - Protection of ecosystems

Culture is sometimes considered the fourth pillar as it shapes human values and practices.

Indicators of Sustainability

Indicators help measure progress towards sustainability:

1. **GDP Growth**
    - Economic growth helps reduce poverty and environmental stress.
2. **Population Stability**
    - Controlled population prevents over-exploitation of resources.
3. **Water Use**

- Availability of clean water for drinking, agriculture and industry.
    - Pollution-free water is essential for health.

4. **Clean Air Index**
    - Measures air quality.
    - Air pollution and greenhouse gases threaten sustainability.
5. **Human Resource Development**
    - Includes education, healthcare, nutrition and living standards.
6. **Energy Intensity**
    - Emphasis on renewable energy.
    - Use of energy should be lower than its regeneration rate.
7. **Soil Degradation**
    - Loss of soil fertility due to erosion threatens agriculture.
8. **Forest Coverage Ratio**
    - Forests regulate climate, water cycle and biodiversity.
    - Deforestation reduces sustainability.
9. **Resource and Material Intensity**
    - Efficient resource use and recycling reduce pressure on nature.
10. **Transport Energy Use**

- Transport sector consumes fossil fuels and causes pollution.

11. **Recycling Proportion**

- Recycling conserves resources and reduces waste.

Q12. India's National Action Plan on Climate Change (NAPCC)

India's National Action Plan on Climate Change (NAPCC)

Introduction

- The **National Action Plan on Climate Change (NAPCC)** was launched in **2008**.
- It was formulated by the **Prime Minister's Council on Climate Change**.
- The plan aims to address the challenges of **climate change through adaptation and mitigation strategies**.
- It creates awareness among **government agencies, scientists, industries, civil society, and communities** about climate risks and solutions.

Objectives of NAPCC

- Protect **poor and vulnerable sections** of society through **inclusive and sustainable development**.
- Achieve **national economic growth** while enhancing **ecological sustainability**.
- Promote **deployment of suitable technologies** for adaptation and mitigation of **greenhouse gas emissions**.
- Encourage **regulatory, developmental, and voluntary mechanisms** for sustainability.
- Ensure **effective implementation** through **public–private partnerships**, civil society, and local governments.
- Promote **international cooperation** in research, technology transfer, and data sharing.

Major Missions of NAPCC

The NAPCC consists of **eight National Missions**:

1. National Solar Mission

- Promotes **solar energy** for power generation and other uses.
- Aims to make solar energy **competitive with fossil fuels**.
- Encourages:
    - Use of **solar thermal technologies** in urban areas and industries.
    - Increase in **photovoltaic production**.
    - Deployment of large-scale **solar power plants**.
- Supports **solar research centres**, domestic manufacturing, and international collaboration.

2. National Mission for Enhanced Energy Efficiency

- Focuses on reducing **energy consumption**.
- Key measures include:
    - Mandatory reduction of energy use in **large industries**.
    - Trading of **energy-saving certificates**.
    - Incentives such as **tax benefits** for energy-efficient appliances.
    - Public–private partnerships for **demand-side energy management**.

3. National Mission on Sustainable Habitat

- Aims to make **urban development energy-efficient**.
- Major initiatives:
    - Expansion of **Energy Conservation Building Code**.
    - Promotion of **waste management and recycling**, including waste-to-energy.
    - Stricter **vehicle fuel efficiency standards**.
    - Encouragement of **public transport systems**.

4. National Water Mission

- Addresses increasing **water scarcity** due to climate change.
- Sets a target of **20% improvement in water-use efficiency**.
- Promotes:
    - Efficient irrigation methods.
    - Water pricing and conservation measures.

5. National Mission for Sustaining the Himalayan Ecosystem

- Focuses on protecting the **Himalayan region**.
- Aims to:
    - Conserve **glaciers**, biodiversity, and forest cover.
    - Monitor climate impacts on **water sources**.
- Important because Himalayan glaciers are a major source of India's rivers.

6. National Mission for a Green India

- Aims to:
    - Afforest **6 million hectares** of degraded forest land.
    - Increase forest cover from **23% to 33%** of India's geographical area.
- Enhances **carbon sinks** and supports biodiversity.

7. National Mission for Sustainable Agriculture

- Focuses on **climate-resilient farming practices**.
- Key strategies:
    - Development of **climate-resilient crop varieties**.
    - Expansion of **crop and weather insurance**.
    - Improved agricultural productivity under changing climate conditions.

8. National Mission on Strategic Knowledge for Climate Change

- Enhances understanding of **climate science and impacts**.
- Promotes:
  - Creation of a **Climate Science Research Fund**.
  - Improved **climate modeling and data systems**.
  - International research collaboration.
  - Private sector participation through **innovation and venture capital**.

Other Supporting Programs under NAPCC

- **Power Generation**:
  - Phasing out inefficient coal-based power plants.
  - Promoting advanced technologies like **supercritical and IGCC**.
- **Renewable Energy**:
  - Mandatory purchase of renewable power under the **Electricity Act, 2003**.
- **Energy Efficiency**:
  - Energy audits for large industries.
  - **Energy labeling** for appliances under the Energy Conservation Act, 2001.

Q13.

(a) Constitutional Provisions Relating to the Environment in India

- India has a long tradition of **environmental protection**, which is reflected in its Constitution.
- The **42nd Constitutional Amendment Act, 1976** gave explicit importance to environmental protection.
- The Constitution provides **directives to the State as well as duties to citizens**.

Important Constitutional Articles

- **Article 48A (Directive Principles of State Policy)**
  - Directs the State to **protect and improve the environment**.
  - Mandates safeguarding of **forests and wildlife**.
- **Article 51A(g) (Fundamental Duties)**
  - It is the **duty of every citizen** to protect and improve the natural environment.
  - Includes protection of **forests, lakes, rivers, wildlife**, and compassion for living creatures.
- **Article 47**
  - Obliges the State to improve **public health**, which is closely linked with environmental quality.
- **Article 253**
  - Empowers Parliament to make laws for implementing **international environmental agreements and conventions**.
- **Article 252**
  - Allows Parliament to legislate on **State List subjects** (like water) if two or more states consent.

Important Environmental Legislations Linked to the Constitution

- **Water (Prevention and Control of Pollution) Act, 1974**
- **Air (Prevention and Control of Pollution) Act, 1981**
- **Environment (Protection) Act, 1986**

Institutional Developments

- **Stockholm Conference, 1972** highlighted global environmental concerns.
- **Tiwari Committee (1980)** recommended environmental governance reforms.
- **Department of Environment** was established as a nodal agency.
- **Ministry of Environment and Forests (1985)** was created for environmental protection.

(b) Wildlife Protection Act, 1972

- The **Wildlife Protection Act, 1972** was enacted to **protect and conserve wildlife** in India.
- It covers **wild animals, birds, and plants**.
- The Act is applicable throughout India (earlier except Jammu & Kashmir).

Objectives of the Act

- Conservation of **biodiversity**.
- Protection of **endangered species**.
- Establishment and management of **national parks, wildlife sanctuaries, and biosphere reserves**.
- Regulation of **hunting and trade** in wildlife products.

Key Provisions

- **Prohibition of hunting** of wild animals except under special circumstances.
- **Chief Wildlife Warden** may permit hunting only if:
  - An animal is dangerous to human life.
  - An animal is diseased beyond recovery.
  - Killing is done in self-defence.

Wildlife Advisory Board

- Constituted in every State/UT.
- Advises the government on:
  - Declaration of protected areas.
  - Wildlife conservation policies.
  - Tribal welfare related to protected areas.

Important Definitions

- **Animal**: Includes mammals, birds, reptiles, and amphibians.
- **Animal Articles**: Items made from animal parts.
- **Hunting Includes**:
  - Killing, capturing, trapping, or poisoning animals.
  - Injuring or destroying any body part of an animal.

(c) Traditional Indian Practices for Nature Conservation

- **Worship of the Sun** as a source of life and energy.
- **Sacred trees** like Peepal and Banyan are protected and worshipped.
- **Animal worship** (cow, snake, elephant, monkey) leading to their conservation.
- **Jain philosophy** promotes non-violence, minimal use of resources, and no pollution.
- **Bishnoi community of Rajasthan** sacrificed lives to protect Khejri trees.
- **Rivers and forests** are considered sacred and protected through religious beliefs.
- **Earth worshipped as "Mother Earth"**, encouraging respect for nature.

Q14. **Explain the main provisions of:**

(a) Water (Prevention and Control of Pollution) Act, 1974

- The Act was enacted to **prevent and control water pollution** and to **restore the wholesomeness of water**.
- It applies to **both public and private sectors**.
- It covers **streams, rivers, lakes, wells, groundwater, and coastal waters**.
- **Central Pollution Control Board (CPCB)** and **State Pollution Control Boards (SPCBs)** were established for implementation.

Functions of the Central Pollution Control Board

- To **coordinate activities** of State Boards and resolve disputes.
- To **advise the Central Government** on matters related to water pollution.
- To **lay down standards** for streams and wells.
- To provide **technical assistance and guidance** to State Boards.
- To promote **research and training** related to water pollution.
- To create **public awareness** regarding water pollution.

Functions of the State Pollution Control Board

- To **prevent, control, and abate water pollution**.
- To **inspect sewage and industrial effluent treatment plants**.
- To **set standards** for discharge of sewage and industrial effluents.
- To monitor water quality in streams and wells.

Important Definitions

- **Pollution**: Contamination or alteration of physical, chemical, or biological properties of water.
- **Outlet**: Any pipe or channel carrying sewage or trade effluent.
- **Sewer**: Pipe or channel carrying sewage or trade effluent.
- **Stream**: Includes rivers, watercourses, inland waters, underground water, and sea or tidal water.

Powers of the Boards

- Power to **obtain information and conduct surveys**.
- Power to **enter and inspect** premises.
- Power to **take samples of water**.
- Power to **prohibit discharge of pollutants** into streams or wells.
- Power to **establish laboratories** and appoint analysts.
- The Act was **amended in 1988** and rules were framed in **1975**.
- Penalties are imposed for **violation of provisions** of the Act.

(b) Air (Prevention and Control of Pollution) Act, 1981

- The Act was enacted to **prevent, control, and abate air pollution**.
- It was introduced in response to concerns raised at the **Stockholm Conference, 1972**.
- The Act aims to **maintain and improve air quality**.
- Central and State Pollution Control Boards were entrusted with enforcement.

Objectives of the Act

- To **establish pollution control boards**.
- To **maintain air quality standards**.
- To **control emissions** from industries and vehicles.

Functions of the Central Pollution Control Board

- To **advise the Central Government** on air pollution matters.
- To **coordinate State Boards** and provide technical assistance.

- To **conduct research and training programmes**.
- To **lay down national air quality standards**.
- To **establish laboratories** for air analysis.

Functions of the State Pollution Control Board

- To **advise State Government** on air pollution control.
- To **inspect air pollution control areas**.
- To **set emission standards** for industries and vehicles.
- To work in coordination with the Central Board.

Powers of the Boards

- State Government may declare any area as an **Air Pollution Control Area**.
- Power to **restrict or prohibit polluting fuels and appliances**.
- Power to **inspect industries and pollution control equipment**.
- Power to **take air samples** from chimneys or atmosphere.
- Power to enforce **emission standards**.

Penalties

- Imprisonment up to **3 months** or **fine up to ₹5,000**, or both.
- Continuing offence attracts an additional **₹100 per day**.

Q15. Differentiate the following:

(a) Wildlife (Protection) Act, 1972 and Forest (Conservation) Act, 1980

Basis	**Wildlife (Protection) Act, 1972**	**Forest (Conservation) Act, 1980**
Objective	Protection of wild animals, birds, and plants	Prevention of deforestation and conservation of forests
Focus	Wildlife conservation	Forest land conservation
Legal Control	Wildlife moved to Concurrent List in 1976	Central Government approval mandatory
Protected Areas	Establishment of sanctuaries, national parks, biosphere reserves	Restricts diversion of forest land for non-forest use
Authorities	Chief Wildlife Warden and Wildlife Boards	Central Government controls forest diversion
Hunting	Generally prohibited with limited exceptions	Not related to hunting

(b) Climate and Weather

Basis	**Climate**	**Weather**
Meaning	Average atmospheric conditions over a long period	Day-to-day condition of the atmosphere
Time Period	Calculated over **30 years or more**	Short-term (hours or days)
Stability	Relatively stable	Changes frequently
Study	Studied under **Climatology**	Studied under **Meteorology**
Elements	Rainfall, humidity, temperature patterns	Rain, storms, heatwaves, snowfall
Nature	Long-term trends	Immediate atmospheric conditions

Q16. Write short notes on:

(a) Environment Protection Act, 1986

- The Environment Protection Act (EPA), 1986 was enacted **after the Bhopal Gas Tragedy (1984)**.
- It is an **umbrella legislation** to protect and improve the environment.
- It provides a framework to **coordinate activities** of authorities under the Water Act and Air Act.
- According to the Act, **environment includes air, water, land**, and their inter-relationship with humans, animals, plants and microorganisms.

Main Features

- Empowers the **Central Government** to take all necessary measures for environmental protection.
- Allows restriction or prohibition of **industries or operations** in specific areas.
- Authorises closure or regulation of polluting industries **without court order**.
- Enables the government to **set emission and effluent standards**.
- Provides power to **collect samples of air, water, soil** for investigation.
- Prescribes **special procedures for handling hazardous substances**.
- Allows **citizens to file complaints** after giving 60 days' notice.
- Provides penalties for violations of the Act.

(b) Forest Conservation Act, 1980

- Forests are vital natural resources for ecological balance.
- The Act was enacted in **1980** to **prevent deforestation**.
- It applies to **all states of India** (except J&K earlier).

Main Provisions

- Restricts use of forest land for **non-forest purposes**.
- Prior approval of **Central Government** is mandatory for diversion of forest land.
- Prevents de-reservation of forests without central approval.
- Establishes a **Forest Advisory Committee**.
- Promotes **compensatory afforestation**.
- Controls **shifting cultivation and encroachment**.
- Penalties imposed for violation of provisions.
- Amended in **1988** to make provisions more stringent.

(c) Nature Reserves

- Nature reserves are **protected areas** set aside to conserve plants, animals or both.
- They are generally **smaller than national parks**.
- The main aim is **protection of endangered species and habitats**.
- Hunting, exploitation and trade of wildlife are strictly controlled.
- They help in **maintaining ecological balance**.
- They provide opportunities for **research and scientific study**.
- Examples include wildlife sanctuaries, biosphere reserves and community reserves.

Examples in India

- Sundarbans Biosphere Reserve (West Bengal)
- Gulf of Mannar Biosphere Reserve (Tamil Nadu)
- Nanda Devi Biosphere Reserve (Uttarakhand)
- Great Nicobar Biosphere Reserve

(d) Smart Cities

- A smart city uses **technology and innovation** to improve quality of life.
- It ensures **sustainable development and efficient service delivery**.
- Focuses on **clean environment and better governance**.

Key Features

- Adequate and safe **water supply**
- Reliable **electricity supply**
- Efficient **solid waste management**
- Sustainable **public transport**
- Affordable housing
- Digital connectivity and **e-governance**
- Environment-friendly infrastructure
- Safety and security of citizens
- Improved **healthcare and education**

(e) Climate Change and Spread of Diseases

- Climate change affects the **spread and intensity of diseases**.
- Many diseases are **climate-sensitive**, such as malaria, dengue and cholera.
- Rising temperatures increase **vector-borne diseases**.
- Heavy rainfall leads to **water contamination** and diarrheal diseases.
- Warmer conditions help pathogens **reproduce faster**.
- Floods and droughts worsen **public health conditions**.
- Climate change increases risks of **epidemics and pandemics**.

(f) Climate Change and Food Security

- Climate change affects **all dimensions of food security**:
    - Food availability
    - Food accessibility
    - Food utilization
    - Food system stability
- Leads to **crop failures**, pest attacks and livestock loss.
- Extreme weather events disrupt **food production and distribution**.
- Vulnerable populations are affected first.
- Coastal, dryland and mountain regions are most at risk.
- Leads to **migration, poverty and conflicts**.
- Agriculture both **contributes to and suffers from climate change**.
- Adaptation and mitigation measures must be integrated into development planning.

(g) 'Odd-Even Formula' and its Impact on Delhi's Air Quality

- The odd-even scheme restricts car use based on **vehicle number plates**.
- Cars with **odd numbers** run on odd dates and **even numbers** on even dates.
- Implemented to reduce **air pollution and traffic congestion**.

Impact

- Significant reduction in **vehicular traffic**.
- Reduction in **fuel consumption**.
- Improvement in **air quality levels**.
- Increased use of **public transport**.
- Raised public awareness about pollution.

(h) Importance of Media for Environmental Issues

- Media plays a crucial role in **environmental awareness**.
- It educates people about **pollution, climate change and conservation**.
- Acts as a **watchdog** over government policies.
- Helps in **public participation** in environmental protection.
- Raises voice against **environmental degradation**.
- Influences policy-makers through public pressure.
- Promotes sustainable practices and environmental ethics.

Q17.

Basis	**National Green Tribunal (NGT)**	**Central Pollution Control Board (CPCB)**
Nature	Judicial body	Statutory administrative body
Year of Establishment	2010	1974
Established Under	NGT Act, 2010	Water Act, 1974
Main Function	Settles environmental disputes	Prevents and controls pollution
Powers	Can pass orders, penalties and compensation	Sets standards and monitors pollution
Role	Delivers environmental justice	Implements pollution control policies

S.No	**Aspect**	**Resettlement**	**Rehabilitation**
1	**Meaning**	Shifting people from their original place due to projects or disasters.	Restoring the living standards, livelihood, and social life of displaced people.
2	**Focus**	Physical relocation to a new place.	Economic, social, and cultural restoration.
3	**Objective**	To provide a new location for affected people.	To ensure normal life and livelihood after displacement.
4	**Scope**	Mainly involves housing and land allotment.	Includes employment, education, health, and community support.
5	**Timeframe**	Done before or during the displacement.	Done after displacement to stabilize affected people.

6	**Examples**	Providing alternate plots of land or houses.	Offering jobs, skill training, financial aid, and social integration programs.

S.No	Aspect	Biosphere Reserve	Biodiversity Hotspot
1	**Meaning**	Protected areas established to conserve ecosystems, species, and genetic diversity along with promoting sustainable development.	Regions rich in biodiversity that are under significant threat of habitat loss.
2	**Objective**	Conservation of species and ecosystems + promoting research, education, and sustainable use of resources.	Focused mainly on the conservation of high biodiversity and endemic species.
3	**Legal Status**	Declared under the **UNESCO Man and Biosphere Programme** and Indian Wildlife (Protection) Act.	Recognized by **Conservation International (CI)** based on criteria of richness and threat.
4	**Structure / Zoning**	Usually divided into **core, buffer, and transition zones**.	No zoning; defined as an area needing urgent conservation.
5	**Coverage**	May include forests, wetlands, grasslands, mountains, etc., covering large areas.	Smaller regions with high species richness and endemism.
6	**Examples in India**	Nilgiri Biosphere Reserve, Sunderban Biosphere Reserve, Nanda Devi Biosphere Reserve.	Himalaya, Indo-Burma, Sundalands, Western Ghats, Indo-Malayan islands.

UNIT - VI

BIODIVERSITY AND CONSERVATION

Q.1. Define Biodiversity. Also explain genetic diversity, species diversity and ecosystem diversity.

OR

Briefly explain the different levels of biodiversity.

Definition of Biodiversity

- Biodiversity refers to the **variety and variability of living organisms** (plants, animals and microorganisms) present in an ecosystem.
- It includes diversity **within species, between species and of ecosystems**.
- The term **"Biodiversity"** was coined by **Raymond F. Dasmann in 1968**.
- Biodiversity represents the **degree of variation of life forms in the biosphere**.
- It plays an important role in **maintaining ecological balance and sustainability**.

Levels of Biodiversity

Biodiversity can be studied at **three major levels**:

(i) Genetic Diversity

- Genetic diversity refers to the **variation in genes within a species**.
- It includes the **total number of genetic characteristics** present in the genetic makeup of a species.
- Genes are responsible for:
  - Physical appearance
  - Biochemical processes
  - Behavioural traits
- Genetic diversity explains why **individuals of the same species differ from each other**.
- A **diverse gene pool** increases the ability of a species to:
  - Adapt to environmental changes
  - Resist diseases
  - Survive adverse conditions
- Genetic diversity is essential for the **healthy survival and evolution of species**.

(ii) Species Diversity

- Species diversity refers to the **number of different species present in a particular area or community**.
- It is measured by:
    - Counting the number of species in a given region
- High species diversity:
    - Prevents inbreeding
    - Maintains ecosystem stability
- **Tropical and subtropical regions** have higher species diversity than temperate regions.
- Undisturbed natural ecosystems have **more species richness** than plantations or modified ecosystems.
- Areas rich in species diversity are known as **"biodiversity hotspots"**.
- The estimated number of species on Earth is about **25–30 million**, while only around **13.9 million** have been identified.

(iii) Ecosystem Diversity

- Ecosystem diversity refers to the **variety of ecosystems found in a particular region**.
- Ecosystems include:
    - Terrestrial ecosystems: forests, grasslands, deserts, mountains
    - Aquatic ecosystems: rivers, lakes, seas, oceans
- Ecosystems may be:
    - Natural (undisturbed)
    - Modified (disturbed by human activities)
- Natural ecosystems support **greater biodiversity** compared to modified ecosystems.
- Ecosystem diversity ensures:
    - Ecological balance
    - Stability of life-support systems

Conclusion

- Biodiversity at all three levels is essential for **environmental sustainability, ecological balance and human survival**.

Q.2. (a) Illustrate the different biogeographic zones of India.

(b) Why is India known as a 'Megadiversity Nation'?

OR

"India is termed as a region of biogeographic zones which impart great diversity to it." Explain.

Introduction

- India ranks **10th among plant-rich countries of the world**.
- Though India occupies only **2% of the world's landmass**, it supports nearly **8% of the world's biodiversity**.
- Countries having very rich biodiversity are known as **Megadiversity Nations**.
- India is divided into **10 biogeographic zones and 26 biotic provinces**, representing almost all major ecosystems of the world.

(a) Biogeographic Zones of India

India is divided into the following **10 biogeographic zones**:

1. Trans-Himalayas

- Extension of the Tibetan Plateau
- Includes **Ladakh and parts of Himachal Pradesh**
- Characterised by **cold deserts**
- Covers about **5.5%** of India's landmass

2. Himalayas

- Extend from **Kashmir to Assam**
- Act as a climatic barrier, preventing cold winds
- Rich in forests, flora and fauna
- Covers about **6.4%** of landmass

3. Desert

- Covers about **6.6%** of landmass
- Includes:
  - Thar Desert (Western Rajasthan)
  - Desert of Gujarat
  - Cold deserts of Jammu & Kashmir and Himachal Pradesh

4. Semi-Arid Zone

- Lies between deserts and Deccan Plateau
- Includes **Aravalli Range**
- Covers about **16.6%** of landmass

5. Western Ghats

- Mountain ranges along the western coast
- One of the **biodiversity hotspots of the world**
- Covers about **4%** of landmass

6. Deccan Plateau

- Large triangular plateau south of the Narmada
- Dominated by deciduous forests
- Covers about **42%** of landmass

7. Gangetic Plain

- Formed by the deposits of River Ganga
- Highly fertile region
- Includes **Sundarbans mangrove forests**
- Covers about **10.8%** of landmass

8. North-East India

- Includes plains and hills of north-eastern states
- Rich vegetation and high rainfall
- Covers about **5.2%** of landmass

9. Islands

- Includes **Andaman and Nicobar Islands**
- About 300 islands, few inhabited
- Covers about **0.3%** of landmass

10. Coasts

- Found along eastern and western boundaries
- Includes **Lakshadweep Islands**
- Covers about **2.5%** of landmass

(b) India as a Megadiversity Nation

- India has **three major biomes**:
  - Tropical humid forests
  - Tropical dry forests
  - Warm deserts

- India's species diversity includes:
  - **350 species of mammals** (8th in the world)
  - **1200 species of birds** (8th)
  - **453 species of reptiles** (5th)
  - **45,000 plant species**, mainly angiosperms
- India has:
  - 1022 species of ferns
  - 1082 species of orchids
  - 50,000 species of insects
- About **18% of Indian plants are endemic**.
- Nearly **50% of lizard species** in India are endemic.
- **Western Ghats and Himalayas** are major centres of biodiversity.
- India has diverse **climatic zones**:
  - Tropical
  - Subtropical
  - Temperate
  - Alpine/Arctic
- These variations create **high biological diversity**.

Conclusion

- Due to its **wide range of ecosystems, species richness and climatic diversity**, India is rightly called a **Megadiversity Nation**.

Q.3. Compare the environmental conditions and biodiversity of different biogeographic zones of India. (2021)

India has **10 major biogeographic zones**, each with **distinct environmental conditions and biodiversity patterns**. A comparative description is given below:

1. Trans-Himalayan Zone

- Located beyond the Greater Himalayas in the **rain-shadow region**
- Covers about **5.6% of India's geographical area**
- Includes **Ladakh, Lahaul–Spiti, Zanskar and parts of Sikkim**
- **Climate**: Cold desert; very low rainfall (less than 350 mm annually)
- **Temperature**: Extremely cold winters; short summers

- **Biodiversity**:
    - Sparse vegetation
    - Alpine grasses, lichens and mosses
    - Fauna: Snow leopard, yak, Tibetan antelope
- **Overall biodiversity**: Low but highly specialised and adapted

2. Himalayan Zone

- Extends from **Jammu & Kashmir to Arunachal Pradesh**
- Covers about **6.41% of India's area**
- Divided into **North-West, West, Central and East Himalayas**
- **Climate**: Varies from subtropical to alpine
- **Rainfall**: 1500 mm (west) to 4800 mm (east)
- **Biodiversity**:
    - Dense forests, alpine meadows
    - High endemism
    - Fauna: Tiger, red panda, musk deer
- **Overall biodiversity**: Very high due to altitudinal variation

3. Indian Desert Zone

- Located in **Western Rajasthan and Rann of Kutch (Gujarat)**
- Covers about **6.5% of landmass**
- **Climate**: Hot and arid
- **Rainfall**: Less than 100 mm in west; up to 500 mm in east
- **Biodiversity**:
    - Sparse thorny vegetation
    - Fauna adapted to drought: Camel, desert fox, reptiles
- **Overall biodiversity**: Low but ecologically specialised

4. Semi-Arid Zone

- Covers **16.6% of India**
- Includes **Punjab, Haryana, Delhi, Eastern Rajasthan, Gujarat**
- **Climate**: Semi-arid with seasonal rainfall
- **Biodiversity**:

  - Grasslands, scrub forests
  - Moderate species diversity
- **Overall biodiversity**: Moderate; transitional ecosystem

5. Western Ghats

- Runs parallel to the **west coast**
- Covers about **4% of landmass**
- **Rainfall**: 2000–7000 mm annually
- **Altitude**: 35 m to 2685 m
- **Biodiversity**:
  - One of the **global biodiversity hotspots**
  - High endemism of plants, amphibians, reptiles
- **Overall biodiversity**: Extremely high

6. Deccan Peninsula

- Largest biogeographic zone (**42% of landmass**)
- Includes central and southern India
- **Climate**: Semi-arid; rain-shadow of Western Ghats
- **Biodiversity**:
  - Tropical deciduous forests
  - Rich fauna: Tiger, elephant, deer
- **Overall biodiversity**: High and widespread

7. Gangetic Plain

- Covers **10.8% of India**
- Formed by **Ganga and Brahmaputra river systems**
- **Climate**: Humid subtropical
- **Biodiversity**:
  - Fertile agricultural land
  - Wetlands, mangroves (Sundarbans)
- **Overall biodiversity**: Moderate but highly productive

8. Coastal Zone

- Coastline length: **7516.6 km**

- Covers **2.5% of landmass**
- Divided into **East Coast, West Coast and Lakshadweep**
- **Biodiversity**:
  - Mangroves, estuaries, coral reefs
  - Rich marine life
- **Overall biodiversity**: High aquatic diversity

9. North-East India

- Covers **5.2% of landmass**
- Includes **Assam valley and North-East hills**
- **Climate**: High rainfall, humid
- **Biodiversity**:
  - Richest plant diversity in India
  - Very high endemism
- **Overall biodiversity**: Exceptionally high

10. Islands

- Includes **Andaman & Nicobar and Lakshadweep**
- Covers **0.3% of landmass**
- **Biodiversity**:
  - Tropical rainforests (Andaman & Nicobar)
  - Coral ecosystems (Lakshadweep)
- **Overall biodiversity**: High endemism, fragile ecosystems

Conclusion

- Environmental conditions such as **climate, rainfall, altitude and geography** directly influence biodiversity.
- Zones like **Western Ghats, Himalayas and North-East India** show highest biodiversity.
- Arid and cold regions have **lower but specialised biodiversity**.

Q.4. Why is India referred to as a Megadiverse Country? (2024 May)

Introduction

- India is one of the **17 megadiverse countries** of the world.
- Although India occupies only **2% of global land area**, it supports nearly **8% of global biodiversity**.

Reasons for India Being a Megadiverse Nation

1. Biogeographic Diversity

- India has **10 biogeographic zones** representing forests, deserts, wetlands, coasts, mountains and islands.
- Located at the **tri-junction of Afro-tropical, Indo-Malayan and Paleo-Arctic realms**.

2. High Species Richness

- Recorded species:
  - 46,000 plant species
  - 81,000 animal species
- Rankings:
  - Mammals – 8th
  - Birds – 8th
  - Reptiles – 5th
  - Angiosperms – 15th

3. High Endemism

- About **4,900 endemic flowering plant species**
- **62% of amphibians** are endemic
- High endemism in:
  - Western Ghats
  - North-East India
  - Himalayas
  - Andaman & Nicobar Islands

4. Ecological Diversity

India has diverse ecosystems such as:

- Forest ecosystems
- Grasslands
- Deserts
- Wetlands
- Freshwater ecosystems
- Marine ecosystems

- Coastal ecosystems

5. Centre of Origin of Cultivated Plants

- One of the **12 centres of origin** of crops
- Home to **167 crop species** and **114 domesticated animal breeds**

6. Conservation Efforts

- Protected Area Network:
  - 85 National Parks
  - 448 Wildlife Sanctuaries
  - Biosphere Reserves
- In-situ and ex-situ conservation:
  - Gene banks
  - Gene sanctuaries (e.g., Tura Range, Meghalaya)

7. Traditional Knowledge Systems

- Ayurveda, Unani, Siddha and Homeopathy depend on biodiversity
- Sustainable use ingrained in Indian culture

Conclusion

- Due to **rich biodiversity, high endemism, ecological variety and strong conservation practices**, India is rightly called a **Megadiverse Nation**.

Q5. Write short notes on

(a) Biodiversity Hotspots

- Biodiversity hotspots are **regions with very high biological diversity** that are **under severe threat due to human activities**.
- The concept of biodiversity hotspots was proposed by **Norman Myers**.
- A region must satisfy **two criteria** to be recognised as a hotspot:
  - It must contain at least **0.5% of the world's endemic vascular plant species**.
  - It must have lost **at least 70% of its original natural vegetation**.
- Hotspots occupy only a small part of Earth's surface but support a large proportion of global biodiversity.
- These areas are important for setting **conservation priorities**.
- Examples of biodiversity hotspots include:
  - **Western Ghats and Sri Lanka**

    - **Himalayas**
    - **Indo-Burma**
    - **Tropical Andes**
- Protection of hotspots helps in saving **endangered and endemic species**.

(b) Endangered Species of India

- Endangered species are species whose **population has declined to a critical level**.
- These species face a **high risk of extinction** in the near future.
- Major causes of endangerment include:
    - Habitat destruction
    - Climate change
    - Pollution
    - Over-exploitation
    - Invasive species
- **Endangered animals of India** include:
    - Bengal Tiger
    - Asian Elephant
    - One-horned Rhinoceros
    - Indian Wild Ass
    - Kashmir Stag (Hangul)
    - Pygmy Hog
- **Endangered birds** include:
    - Great Indian Bustard
    - Siberian Crane
    - Florican
    - Vultures
- **Endangered reptiles and amphibians**:
    - Gharial
    - Flying frog
- **Endangered plants**:
    - Orchids and medicinal plants threatened due to overharvesting

(c) Endemic Species of India

- Endemic species are species that are **found only in a specific geographical area**.
- India has a high number of endemic species due to its **diverse climate and geography**.
- Nearly **60% of India's endemic species** are found in:
    - Western Ghats
    - Himalayas
- Major regions of endemism in India:
    - North-East India
    - Western Ghats
    - North-West Himalayas
    - Andaman and Nicobar Islands
- **Endemic flora examples**:
    - Nepenthes khasiana
    - Sapria himalayana
- **Endemic fauna examples**:
    - Lion-tailed macaque
    - Nilgiri langur
    - Brown palm civet
- Threats to endemic species include:
    - Habitat loss
    - Pollution
    - Diseases
    - Climate change

(d) Biodiversity

- Biodiversity refers to the **variety and variability of living organisms** on Earth.
- It includes plants, animals, microorganisms and ecosystems.
- Biodiversity is essential for **ecological balance and survival of life**.
- There are **three levels of biodiversity**:
    - **Genetic diversity** – variation of genes within a species
    - **Species diversity** – variety of species in a region

    - **Ecosystem diversity** – variety of ecosystems and habitats
- Biodiversity provides:
    - Food and medicine
    - Ecological stability
    - Cultural and economic benefits

(e) Keystone Species

- Keystone species are species that have a **disproportionately large effect** on their ecosystem.
- Their removal can cause **major changes or collapse of the ecosystem**.
- They may be **few in number**, but their ecological role is critical.
- Keystone species help maintain:
    - Species diversity
    - Food chains
    - Ecosystem structure
- Examples of keystone species:
    - Wolf
    - Otter
    - Prairie dog
    - Bison

(f) Flagship Species

- Flagship species are **charismatic species used as symbols** for conservation campaigns.
- They help **raise public awareness and support** for conservation.
- Protection of flagship species indirectly protects their habitats.
- These species are chosen because of their **popular appeal**.
- Examples:
    - Bengal Tiger (India)
    - Giant Panda (China)
- Flagship species play an important role in **policy making and conservation planning**.

(g) Umbrella Species

- Umbrella species are species whose **conservation automatically protects many other species**.

- These species usually require **large habitats**.
- By protecting their habitat, the entire ecosystem is conserved.
- Umbrella species are useful in:
    - Designing protected areas
    - Planning wildlife reserves
- Example:
    - **Tiger** – conserving tigers helps protect forests and other wildlife like deer, birds and reptiles

(h) Indicator Species

- Indicator species are organisms that **reflect the health of an ecosystem**.
- Their presence, absence or population size indicates **environmental conditions**.
- They help in monitoring:
    - Pollution levels
    - Habitat degradation
- Examples:
    - Lichens – indicate air pollution
    - Tubifex worms – indicate polluted, oxygen-poor water
    - Frogs – indicate water and soil pollution
- Indicator species are useful tools for **environmental assessment and management**.

(i) Islands of India

- India has **more than 1000 islands**, grouped mainly into two categories:
    - Andaman and Nicobar Islands
    - Lakshadweep Islands

Andaman and Nicobar Islands

- Located in the **Bay of Bengal**
- Formed by submerged mountain ranges
- Have **tropical rainforests**
- Rich in biodiversity and endemic species
- Some islands have **volcanic origin**
- Coral reefs present around many islands

Lakshadweep Islands

- Located in the **Arabian Sea**
- Formed by **coral deposits**
- Smaller and closer to mainland India
- Soil is fertile and suitable for coconut cultivation
- Have rich marine biodiversity

Q.6. Explain the IUCN Red List Criteria and Categories in detail and discuss their significance in assessing the conservation status of plant and animal species.

Introduction

- The **IUCN Red List of Threatened Species** is developed by the **International Union for Conservation of Nature (IUCN)**.
- It is the **most comprehensive and globally accepted system** for assessing the **risk of extinction** faced by plant and animal species.
- The Red List helps in **evaluating, monitoring and conserving biodiversity** by classifying species based on scientific criteria.

IUCN Red List Criteria

- The IUCN Red List Criteria were first introduced in **1994** and revised periodically.
- These criteria are **objective, quantitative and science-based**.
- Species are evaluated using data from:
    - Field surveys
    - Scientific research
    - Expert opinions
    - Museum records
    - Published literature
- The assessment is based on **five major criteria**:

1. Population Size

- Assesses the **total number of mature individuals** in a species.
- Small population sizes increase vulnerability to extinction due to:
    - Inbreeding
    - Environmental changes
    - Human disturbances

2. Geographic Range

- Evaluates the **extent of occurrence** and **area of occupancy**.
- Species with a **restricted or fragmented range** are more vulnerable.

3. Population Decline

- Measures the **rate of decline** in population over a specific time period.
- Decline may be due to:
  - Habitat loss
  - Climate change
  - Overexploitation
  - Pollution
  - Diseases

4. Population Fragmentation

- Assesses whether populations are **isolated or fragmented**.
- Fragmentation reduces:
  - Genetic diversity
  - Reproductive success
  - Long-term survival chances

5. Probability of Extinction

- Estimates the **likelihood of extinction** within a defined time frame (often 100 years).
- Uses population models and threat analysis.

IUCN Red List Categories

Based on the above criteria, species are placed into the following categories:

1. Least Concern (LC)

- Species with **stable and widespread populations**.
- No immediate threat of extinction.

2. Near Threatened (NT)

- Species close to qualifying for a threatened category.
- May become threatened if pressures continue.

3. Vulnerable (VU)

- Species facing a **high risk of extinction** in the medium term.

- Significant population decline or habitat loss observed.

4. Endangered (EN)

- Species facing a **very high risk of extinction** in the near future.
- Population size is severely reduced.

5. Critically Endangered (CR)

- Species facing an **extremely high risk of extinction**.
- Very small population or extremely restricted range.

6. Extinct in the Wild (EW)

- Species surviving **only in captivity or cultivation**.
- No known populations in natural habitats.

7. Extinct (EX)

- Species with **no surviving individuals**.

8. Data Deficient (DD)

- Insufficient data available to assess extinction risk.
- Does not mean species is not threatened.

9. Not Evaluated (NE)

- Species that have **not yet been assessed**.

Significance of the IUCN Red List

1. Conservation Prioritization

- Helps identify **species at greatest risk**.
- Guides allocation of **limited conservation resources**.

2. Policy and Legal Framework

- Used by governments to frame:
  - Wildlife protection laws
  - Biodiversity action plans
- Many species receive **legal protection** based on Red List status.

3. Monitoring and Trend Analysis

- Enables **long-term monitoring** of species populations.
- Helps evaluate effectiveness of conservation strategies.

4. Public Awareness and Education

- Raises awareness about:
    - Threatened species
    - Importance of biodiversity conservation
- Encourages public participation and support.

5. International Cooperation

- Provides a **common global standard**.
- Facilitates international research, funding and conservation programs.

Conclusion

- The IUCN Red List Criteria and Categories form a **scientifically reliable and standardized system** for assessing extinction risk.
- By analyzing population size, geographic range, decline, fragmentation and extinction probability, the Red List provides a **holistic view of species conservation status**.
- It plays a vital role in **biodiversity conservation, policy formulation and global environmental management**.
- Effective use of the IUCN Red List is essential for ensuring the **long-term survival of plant and animal species** on Earth.

Q.7. Discuss the importance of the Himalaya as a Global Biodiversity Hotspot and an ecologically fragile area.

Introduction

- The **Himalayas** are one of the **most important global biodiversity hotspots**.
- They are recognized as a hotspot due to:
    - **High species richness**
    - **High endemism**
    - **Severe threats from human activities**
- The region is also considered **ecologically fragile** because of its sensitive climate, young geology, and increasing anthropogenic pressures.

Overview of the Himalaya Hotspot

- The Himalaya hotspot stretches over **3,000 km** across:
    - Northern Pakistan
    - India (Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh)
    - Nepal
    - Bhutan

    - Parts of China (Tibet) and Myanmar
- Covers an area of about **750,000 km²**.
- Includes all mountain peaks above **8,000 metres**, including **Mt. Everest (Sagarmatha)**.
- Divided into:
    - **Eastern Himalaya**
    - **Western Himalaya**
- The **Kali Gandaki River gorge** acts as a major biological barrier influencing species distribution.

Diversity of Ecosystems

- The abrupt rise in altitude from **below 500 m to above 8,000 m** results in a wide range of ecosystems within short distances.
- Major ecosystems include:
    - Alluvial grasslands (among the tallest in the world)
    - Subtropical broadleaf forests
    - Temperate forests
    - Coniferous forests
    - Alpine meadows
    - Permanent snow and ice zones
- This vertical zonation supports **extraordinary ecological diversity**.

Species Diversity and Biogeographic Importance

- The Himalayas lie at the junction of:
    - **Palearctic realm**
    - **Indo-Malayan realm**
- Species from both realms coexist, increasing biodiversity.
- Geological diversity, varied climate, altitude differences and complex topography contribute to high species richness.

Plant Diversity and Endemism

- About **10,000 plant species** are found in the Himalaya hotspot.
- Nearly **3,160 species are endemic**.
- **71 endemic genera** are present.
- **Five plant families** are endemic to the region.

- The largest flowering plant family is **Orchidaceae** with around **750 species**.
- Eastern Himalaya is a centre of diversity for:
    - *Rhododendron*
    - *Primula*
    - *Pedicularis*
- Vascular plants have been recorded at altitudes above **6,000 m**, indicating exceptional adaptability.

Faunal Diversity

- The Himalayas support important populations of large mammals and birds such as:
    - Tiger
    - Snow leopard
    - Asian elephant
    - One-horned rhinoceros
    - Red panda
    - Wild water buffalo
    - Himalayan vultures
- Many species are **endangered or threatened**, increasing conservation importance.

Ecological Fragility of the Himalayas

- The Himalayas are **geologically young and unstable**, making them prone to:
    - Landslides
    - Earthquakes
    - Soil erosion
- The region is highly sensitive to:
    - Climate change
    - Glacial melting
    - Altered rainfall patterns
- Even small disturbances can cause **large-scale ecological damage**.

Major Threats to Himalayan Biodiversity

1. Habitat Loss and Deforestation

- Conversion of forests into:

    - Agricultural land
    - Human settlements
- Large-scale deforestation in:
    - Nepal
    - Sikkim
    - Darjeeling
    - Assam

2. Agriculture and Grazing Pressure

- Cultivation at high altitudes of crops like:
    - Barley
    - Potato
    - Buckwheat
- Overgrazing by cattle and yak damages alpine meadows.
- Use of fire for clearing land increases forest fire risks.

3. Overexploitation of Resources

- Excessive extraction of:
    - Fuelwood
    - Timber
    - Medicinal plants
- Medicinal plant collectors often uproot entire plants, preventing regeneration.

4. Poaching and Illegal Wildlife Trade

- Tigers and rhinoceros hunted for traditional medicine.
- Snow leopard and red panda hunted for fur.
- Serious threat to already endangered species.

5. Developmental Activities

- Construction of:
    - Roads
    - Dams
    - Mining projects
- Leads to habitat fragmentation and ecological imbalance.

6. Tourism and Pollution

- Unplanned tourism causes:
    - Waste accumulation
    - Water pollution
    - Habitat disturbance
- Use of agrochemicals pollutes soil and water bodies.

Conclusion

- The Himalayas are a **global biodiversity hotspot** due to their rich ecosystems, high endemism and unique species.
- At the same time, they are **ecologically fragile**, highly sensitive to human and climatic disturbances.
- Conservation of the Himalayas is essential for:
    - Biodiversity protection
    - Climate regulation
    - Water security for millions of people
- Sustainable development, strict conservation measures and community participation are crucial to protect this invaluable ecosystem.

Q8. Illustrate the importance of 'Hotspots' in conserving biodiversity.

Compare ecosystems of the Himalayan Hotspot and Western Ghats Hotspot in a tabular format.

Importance of Biodiversity Hotspots in Conserving Biodiversity

- **Biodiversity hotspots** are regions with exceptionally high species richness and endemism, but which are also under severe threat.
- Protecting hotspots ensures **maximum conservation benefits with limited resources**.

Key Importance of Hotspots

- **Conservation of endemic species**
    - Hotspots contain species found nowhere else on Earth.
    - Loss of a hotspot means permanent global extinction of endemic species.
- **Maintenance of ecosystem stability**
    - Biodiversity-rich ecosystems are more resilient to disturbances.
    - They help maintain ecological balance.
- **Provision of ecosystem services**

- Hotspots contribute significantly to:
    - Clean air and water
    - Climate regulation
    - Soil fertility
    - Carbon sequestration

- **Support to human livelihoods**
  - Many indigenous and local communities depend directly on hotspot ecosystems.
  - Hotspots provide food, medicine, fuel and shelter.
- **Climate regulation**
  - Forests and oceans within hotspots absorb carbon dioxide.
  - They help in moderating global and regional climate.
- **Protection of vulnerable populations**
  - Hotspots overlap with regions supporting vulnerable human populations.
  - These areas provide about **35% of ecosystem services** despite covering only **~25% of Earth's land surface**.
- **Conservation prioritization**
  - Mapping hotspots helps policymakers identify priority areas for conservation investment.

Comparative Study of Ecosystems in Himalayan Hotspot and Western Ghats Hotspot

Table: Himalayan Hotspot vs Western Ghats Hotspot

Basis of Comparison	**Himalayan Hotspot**	**Western Ghats Hotspot**
Geographical location	Northern India, Nepal, Bhutan, Pakistan, China	Western coastal region of India
Altitude range	<500 m to >8,000 m	300 m to ~2,700 m
Climate (Abiotic)	Extreme variation: tropical to alpine; cold temperatures at high altitudes	Tropical monsoon climate; warm and humid
Rainfall (Abiotic)	Varies greatly; higher in Eastern Himalayas	High rainfall due to southwest monsoon
Major ecosystems	Alluvial grasslands, subtropical forests, temperate forests, coniferous forests, alpine meadows	Tropical evergreen forests, moist deciduous forests, shola grasslands, wetlands

Soil type (Abiotic)	Thin, immature soils; prone to erosion	Lateritic and forest soils; rich in organic matter
Vegetation (Biotic)	Rhododendrons, orchids, conifers, alpine herbs	Dipterocarpus, Impatiens, Calamus, orchids, ferns
Plant endemism	About 3,160 endemic plant species	58 endemic plant genera
Major plant families	Tetracentraceae, Hamamelidaceae, Orchidaceae	Dipterocarpaceae, Orchidaceae, Rubiaceae
Faunal diversity (Biotic)	Snow leopard, tiger, red panda, rhino, wild buffalo, vultures	Asian elephant, tiger, lion-tailed macaque, amphibians
Freshwater biodiversity	Rivers with cold-water species	Very high freshwater fish endemism (140+ species)
Amphibian and reptile diversity	Moderate but specialized cold-adapted species	Extremely high reptile and amphibian endemism
Human pressure	Agriculture, grazing, deforestation, dams	Deforestation, plantations, mining, urbanisation
Ecological fragility	Very high due to young geology and landslides	High due to habitat fragmentation and deforestation

Conclusion

- Biodiversity hotspots are crucial for **global conservation efforts** as they safeguard rare, endemic and threatened species.
- The **Himalayan hotspot** is characterized by extreme altitudinal variation and alpine ecosystems, while the **Western Ghats hotspot** is known for tropical forests and high endemic amphibian and fish diversity.
- Protecting both hotspots is essential not only for biodiversity conservation but also for **climate regulation, ecosystem services and human well-being**.
- Sustainable development and strict conservation policies are necessary to preserve these ecologically sensitive regions.

Q.9. Major Threats to Biodiversity

- Biodiversity is under severe threat mainly due to **human activities** such as population growth, over-exploitation of natural resources, industrialisation, urbanisation, pollution and climate change. The major threats to biodiversity are explained below:

(a) Habitat Loss

- Habitat loss is the **primary and most serious threat** to biodiversity.
- It refers to the **destruction, degradation or fragmentation** of natural habitats.

- Major causes include:
    - Deforestation for agriculture, settlements and industries
    - Construction of dams, roads and reservoirs
    - Mining and urban expansion
    - Draining of wetlands

Effects of Habitat Loss

- Loss of shelter, food and breeding grounds for species
- Fragmentation isolates populations, reducing genetic diversity
- Difficulty in finding mates and food
- Increased vulnerability to extinction
- Disturbance of food chains and ecosystems

(b) Poaching of Wildlife

- Poaching is the **illegal hunting, killing or capturing of wild animals**.
- Animals are poached for:
    - Meat
    - Skins, hides and fur
    - Ivory, horns, bones and teeth
    - Traditional medicines
    - Illegal wildlife trade or thrill hunting

Effects of Poaching

- Rapid decline of wildlife populations
- Disturbance of food webs and ecological balance
- Loss of keystone and apex species
- Spread of diseases due to consumption of wild meat
- Examples:
    - Tiger, rhinoceros, elephant, pangolin
    - Ebola outbreaks linked to bush-meat consumption

(c) Man–Wildlife ConflictS

- Man–wildlife conflict occurs when **humans and wild animals compete for space and resources**.

- It is increasing due to:
    - Shrinking wildlife habitats
    - Encroachment into forest areas
    - Depletion of natural prey
    - Expansion of agriculture near forests

Forms of Man–Wildlife Conflict

- Crop raiding by elephants, wild boars and deer
- Attacks on livestock by tigers and leopards
- Human injuries and deaths
- Property damage

Significance in India

- Common examples:
    - Elephant–human conflict
    - Tiger and leopard attacks
    - Monkey menace in urban areas
- Leads to:
    - Retaliatory killing of animals
    - Decline in wildlife populations
    - Loss to farmers and local communities
    - Negative attitude towards conservation

(d) Biological Invasions

- Biological invasion refers to the **introduction of non-native species** into a new region where they spread aggressively.
- These species lack natural predators in the new environment.

Causes

- Trade and transport
- Agriculture and horticulture
- Tourism and globalisation
- Disturbance of natural ecosystems

Impacts of Biological Invasions

- Outcompete native species for food and space
- Alter habitats and ecosystem functioning
- Predation on native species
- Spread of diseases and pathogens
- Hybridisation causing loss of genetic purity

Examples

- Water hyacinth
- Lantana camara
- Parthenium
- African catfish

(e) Excessive Pollution

- Pollution adversely affects **air, water, soil and living organisms**.
- Major sources include:
    - Industries
    - Vehicles
    - Power plants
    - Agricultural chemicals
    - Plastic waste

Effects on Biodiversity

- Air pollution affects plant photosynthesis and animal health
- Water pollution kills aquatic organisms
- Soil pollution reduces fertility and microbial diversity
- Bioaccumulation and biomagnification of toxins
- Climate change due to greenhouse gases
- Coral bleaching and species extinction

Q10. **Role of Human Activities as a Threat to Biodiversity Hotspots in India**

India is part of **four global biodiversity hotspots**:

- Himalaya
- Western Ghats
- Indo-Burma

- Sundaland (Andaman & Nicobar Islands)

Human Activities Threatening Hotspots

- **Deforestation**
  - Clearing forests for agriculture, plantations and settlements
  - Major threat in Western Ghats and North-East India
- **Urbanisation and infrastructure development**
  - Roads, dams, hydropower projects
  - Causes habitat fragmentation in Himalayas
- **Mining and quarrying**
  - Leads to habitat destruction and pollution
  - Common in Western Ghats and Eastern Himalayas
- **Over-exploitation of resources**
  - Overharvesting of timber, medicinal plants and fuelwood
  - Illegal wildlife trade
- **Tourism pressure**
  - Unplanned tourism causes waste generation and habitat disturbance
- **Agricultural expansion**
  - Shifting cultivation and monoculture plantations
  - Loss of native species
- **Climate change**
  - Alters temperature and rainfall patterns
  - Threatens endemic and sensitive species

Conclusion

- Biodiversity loss is largely **human-induced**.
- Habitat loss, poaching, invasive species, pollution and conflicts are major drivers.
- Conservation of biodiversity hotspots is crucial for:
  - Protecting endemic species
  - Maintaining ecosystem services
  - Ensuring sustainable development
- Strong laws, public awareness and sustainable practices are essential.

Q11. **Human–Wildlife Conflict (HWC)**

Meaning / Definition

- Human–Wildlife Conflict (HWC) refers to any interaction between humans and wildlife that results in:
    - Loss of human life or injury
    - Damage to crops, livestock or property
    - Threat to wildlife survival and conservation
- It occurs when human activities negatively affect wildlife or when wildlife adversely affects human interests.
- HWC is a serious challenge for biodiversity conservation, especially in developing countries like India.

Reasons / Causes of Human–Wildlife Conflict

1. Shrinking Wildlife Habitat

- Rapid population growth has led to expansion of human settlements.
- Forest land is cleared for agriculture, industries, roads and housing.
- Encroachment into forest areas reduces living space for wild animals.
- Animals are forced to move into human-dominated areas in search of food and shelter.

2. Loss of Natural Food and Water Resources

- Overuse of forest resources by humans reduces food availability for animals.
- Grazing of livestock in forests competes with wild herbivores.
- Drying of water bodies forces animals to move towards villages.

3. Decline in Prey Population

- Poaching and habitat loss reduce prey species.
- Carnivores like tigers and leopards attack livestock and humans due to lack of prey.

4. Infrastructure Development

- Roads, railways and dams pass through wildlife corridors.
- Leads to frequent animal deaths due to vehicle and train collisions.

Examples of Human–Wildlife Conflict in India

1. Elephant–Human Conflict

- Seen in Bandipur Tiger Reserve, Nilgiri Biosphere Reserve and Anamalai Tiger Reserve.
- Elephants destroy crops, houses and sometimes kill humans.

- Retaliatory killings and injuries to elephants are common.

2. Big Cat Conflicts

- Leopards frequently attack humans in Maharashtra, Uttarakhand and Himachal Pradesh.
- Snow leopard conflicts are common in the Himalayan region.
- Tigers kill cattle and occasionally humans near forest fringes.

3. Crop Raiding by Herbivores

- In Sariska Wildlife Sanctuary, Nilgai, Sambar and Chital damage crops.
- Farmers suffer heavy economic losses.

4. Monkey Menace

- In urban areas like Vrindavan (Uttar Pradesh), rhesus monkeys:
  - Bite people
  - Steal food
  - Damage houses and public property

5. Transport-Related Conflicts

- Wildlife accidents on highways and railway tracks are common.
- Animals get injured or killed due to fast-moving vehicles.

Measures to Prevent or Reduce Human–Wildlife Conflict

1. Wildlife Translocation or Human Relocation

- Relocating animals or human settlements from conflict-prone areas.
- Creation of buffer zones around protected areas.

2. Electric Fencing and Barriers

- Use of solar-powered electric fencing around farms and villages.
- Prevents animals from entering crop fields.

3. Habitat Conservation and Restoration

- Protecting forests and wildlife corridors.
- Restoring degraded habitats to ensure food and water availability.

4. Public Awareness and Education

- Educating people about wildlife behaviour and safety measures.
- Promoting coexistence with wildlife.

5. Community Participation

- Involving local communities in wildlife management.
- Compensation schemes for crop and livestock losses.

6. Traffic Control Measures

- Speed limits and animal crossings on highways.
- Construction of underpasses and overpasses for animals.

Biological Invasion and Its Impact on Indian Biodiversity

Meaning of Biological Invasion

- Biological invasion refers to the introduction of non-native (alien) species into an ecosystem.
- These species spread rapidly and cause harm to native plants, animals and ecosystems.
- India has been highly affected due to historical and modern human activities.

Impact of Biological Invasion on Biodiversity

- Invasive species outcompete native species for food, water and space.
- They alter ecosystem functions like:
    - Pollination
    - Seed dispersal
    - Nutrient cycling
- Native species may decline or become extinct.
- Overall biodiversity and ecological balance are disturbed.

Major Invasive Species in India

1. Water Hyacinth (Eichhornia crassipes)

- Aquatic plant growing rapidly in stagnant water bodies.
- Forms dense mats blocking sunlight and oxygen.
- Causes death of aquatic organisms.
- Obstructs irrigation, navigation and water supply.

2. Parthenium (Congress Grass)

- Aggressive weed invading agricultural fields.
- Releases toxic chemicals affecting crop growth.
- Causes skin allergies, asthma and health problems.
- Harmful to livestock.

3. Lantana Camara

- Forms dense thickets in forest areas.
- Prevents growth of native plants.
- Reduces food availability for herbivores.
- Disturbs forest regeneration.

4. Prosopis Juliflora (Vilayati Babool)

- Displaces native vegetation.
- Reduces biodiversity.
- Consumes large amounts of groundwater.
- Forms dense thorny thickets in dry regions.

Measures to Control Biological Invasion

1. Prevention

- Strict biosecurity checks at ports, airports and borders.
- Laws restricting import and sale of invasive species.
- Public awareness about invasive species.

2. Early Detection and Rapid Response (EDRR)

- Regular monitoring of ecosystems.
- Public reporting systems for invasive species sightings.
- Immediate eradication in early stages.

3. Integrated Pest Management (IPM)

- **Biological Control:** Use of natural predators.
- **Mechanical Control:** Manual removal, mowing or trapping.
- **Chemical Control:** Careful use of pesticides and herbicides.

4. Habitat Restoration

- Planting native species.
- Restoring wetlands and degraded ecosystems.
- Preventing soil erosion and further spread of invasives.

Q.12. Mass Extinction Crisis: Causes, Consequences and Solutions

Introduction

- The current mass extinction crisis refers to the rapid and widespread loss of species across the Earth.
- Scientists consider it the **sixth mass extinction**, primarily driven by **human activities**.
- This crisis poses a serious threat to **global biodiversity, ecosystem stability and human well-being**.
- Understanding its causes, consequences and solutions is essential to mitigate its long-term impacts.

Causes of the Current Mass Extinction Crisis

1. Habitat Destruction

- Habitat loss is the **leading cause** of species extinction.
- Activities such as:
    - Deforestation
    - Urbanization
    - Industrial expansion
    - Agricultural land conversion
      destroy natural ecosystems.
- Example:
    - Expansion of **palm oil plantations in Southeast Asia** has destroyed rainforests.
    - Species like **orangutans** have lost their natural habitats and are critically endangered

2. Climate Change

- Human-induced climate change is caused by greenhouse gas emissions.
- Rising temperatures lead to:
    - Melting glaciers
    - Sea level rise
    - Extreme weather events
- Many species cannot adapt quickly to changing conditions.
- Example:
    - **Coral reefs** experience bleaching due to warmer oceans, threatening marine biodiversity.

3. Pollution

- Pollution contaminates air, water and soil.
- Sources include:

    - Industrial waste
    - Agricultural chemicals
    - Plastic pollution
- Pollutants disrupt food chains and ecosystem processes.
- Example:
    - **Great Pacific Garbage Patch** harms marine animals and seabirds through ingestion and entanglement.

4. Invasive Species

- Non-native species introduced intentionally or accidentally disturb local ecosystems.
- They:
    - Outcompete native species
    - Spread diseases
    - Alter habitats
- Example:
    - **Cane toad in Australia** caused decline of native predators.

5. Overexploitation of Resources

- Excessive hunting, fishing and harvesting reduce species populations.
- Illegal wildlife trade increases extinction risk.
- Example:
    - Poaching for **ivory, rhino horns and shark fins** has endangered many species.

Consequences of the Mass Extinction Crisis

1. Loss of Biodiversity

- Rapid extinction reduces species richness and genetic diversity.
- Ecosystems become less resilient to environmental changes.

2. Disruption of Ecosystems

- Loss of key species destabilizes ecosystems.
- Leads to:
    - Increase in pests and diseases
    - Reduced ecosystem productivity
- Example:

    - Loss of forests contributes to **floods, droughts and wildfires**.

3. Economic Consequences

- Biodiversity loss affects:
    - Agriculture
    - Fisheries
    - Medicine
    - Tourism
- Example:
    - Decline of **pollinators like bees** threatens global food security.

4. Ethical and Moral Implications

- Raises ethical questions about human responsibility.
- Every species has **intrinsic value** and the right to exist.
- Example:
    - Construction of resorts near forests destroys habitat of endangered birds.

5. Impact on Human Well-being

- Biodiversity provides essential ecosystem services such as:
    - Clean air and water
    - Disease regulation
    - Food security
- Loss of biodiversity increases risks of:
    - Diseases
    - Malnutrition
    - Environmental disasters

Potential Solutions to the Mass Extinction Crisis

1. Habitat Conservation

- Protection and restoration of natural habitats.
- Creation of:
    - National parks
    - Wildlife sanctuaries
    - Marine protected areas

2. Sustainable Land Use and Agriculture

- Adoption of:
    - Reforestation
    - Agroforestry
    - Organic farming
- Reduced use of harmful pesticides and fertilizers.

3. Climate Change Mitigation

- Reduction of greenhouse gas emissions.
- Promotion of:
    - Renewable energy
    - Energy efficiency
    - Climate-friendly policies

4. Pollution Control

- Strong environmental regulations.
- Reduction of plastic use.
- Improved waste management and recycling systems.

5. Invasive Species Management

- Prevention through strict biosecurity measures.
- Early detection and rapid response.
- Controlled eradication and monitoring programs.

6. Sustainable Resource Management

- Regulated fishing and hunting practices.
- Strict enforcement against illegal wildlife trade.
- Community-based conservation programs.

Ethical, Economic and Environmental Implications

Ethical Implications

- Moral duty to protect nature and other species.
- Responsibility towards future generations.
- Recognition of rights of non-human life forms.

Economic Implications

- Conservation ensures long-term economic benefits.
- Sustainable use of resources supports livelihoods.
- Prevents future economic losses from ecosystem collapse.

Environmental Implications

- Protecting biodiversity strengthens ecosystem services.
- Helps mitigate climate change.
- Ensures ecological balance and planetary health.

Q.13. Major Threats to Biodiversity and Measures to Mitigate Them

Major Threats to Biodiversity

Biodiversity across the world is declining rapidly due to various human-induced activities. The major threats are as follows:

1. Habitat Loss and Fragmentation

- Destruction of natural habitats due to deforestation, urbanization, mining and agricultural expansion.
- Fragmentation isolates populations, reduces breeding opportunities and increases extinction risk.

2. Overexploitation of Species

- Excessive hunting, fishing, logging and harvesting of plants beyond sustainable limits.
- Leads to population decline and extinction of species.

3. Invasive Alien Species

- Introduction of non-native species that outcompete, prey upon or bring diseases to native species.
- Alters ecosystem structure and functioning.

4. Climate Change

- Global warming, rising sea levels and extreme weather events disrupt ecosystems.
- Species unable to adapt migrate or face extinction.

5. Pollution

- Air, water and soil pollution from industries, agriculture and plastic waste.
- Toxic substances accumulate in food chains causing bio-magnification.

Measures to Mitigate Three Major Threats at the Global Level

1. Protecting Natural Areas (Mitigating Habitat Loss)

- Establishment of **protected areas** such as national parks, wildlife sanctuaries and biosphere reserves.
- Restricting deforestation, mining and human interference in ecologically sensitive regions.
- Proper planning to include:
    - Breeding sites
    - Feeding areas
    - Water sources
- Well-designed protected areas conserve **entire ecosystems**, not just individual species.

2. Preventing Species Introductions (Mitigating Invasive Species Threat)

- Prevention is more effective and economical than eradication.
- Governments enforce **strict biosecurity laws** to restrict the introduction of foreign plants and animals.
- Inspection of cargo, ships and luggage at borders.
- Public awareness about dangers of releasing exotic pets or plants into the wild.

3. Slowing Climate Change (Mitigating Climate Threats)

- Reduction of greenhouse gas emissions by:
    - Promoting renewable energy
    - Improving energy efficiency
    - Reducing fossil fuel use
- International agreements like the **Paris Agreement** help control global warming.
- Climate mitigation reduces species migration stress and ecosystem disruption.

Additional Supporting Measures

4. Education and Awareness

- Educating people about biodiversity loss and conservation.
- Encouraging sustainable lifestyles and eco-friendly practices.

5. Promoting Sustainability

- Adoption of sustainable agriculture, forestry and fishing practices.
- Prevents overexploitation and habitat destruction.
- Maintains long-term ecological balance.

Q.14. Conservation of Biodiversity

Meaning of Conservation of Biodiversity

- Conservation refers to the **protection, preservation, management and restoration** of biodiversity and natural resources.
- It aims to prevent extinction, maintain ecosystem balance and ensure sustainable use of biological resources.
- Conservation focuses on protecting:
    - Species
    - Habitats
    - Genetic diversity

Methods of Conservation

Biodiversity conservation can be achieved through **two main methods**:

I. In-Situ Conservation

Meaning

- Conservation of species **within their natural habitats**.
- Maintains ecological interactions and evolutionary processes.

Methods of In-Situ Conservation

1. Biosphere Reserves

- Large protected areas promoting conservation along with sustainable use.
- India has **18 biosphere reserves**.
- Divided into three zones:
    - **Core Zone**: No human activity allowed.
    - **Buffer Zone**: Limited research, tourism and education.
    - **Transition Zone**: Sustainable human activities allowed.

2. National Parks

- Protected by central or state legislation.
- No human activities like grazing or cultivation allowed.
- Preserve natural habitats and gene pools.
- Examples:
    - Jim Corbett National Park (Uttarakhand)
    - Kaziranga National Park (Assam)

3. Wildlife Sanctuaries

- Created by state or central government orders.

- Limited human activities allowed with permission.
- Examples:
  - Mhadei Wildlife Sanctuary (Goa)
  - Bondla Wildlife Sanctuary (Goa)

II. Ex-Situ Conservation

Meaning

- Conservation of biodiversity **outside natural habitats**.
- Used when species are critically endangered.

Methods of Ex-Situ Conservation

1. Zoos

- Animals kept in controlled environments.
- Promote captive breeding of endangered species.

2. Captive Breeding

- Artificial breeding to increase population size.
- Later reintroduced into the wild.

3. Botanical Gardens

- Conservation of plant species for research and education.
- Include herbariums, seed collections and laboratories.

4. Gene Banks

- Storage of genetic material such as seeds, DNA and tissues.
- Cryopreservation ensures long-term genetic conservation.

Difference Between In-Situ and Ex-Situ Conservation

In-Situ Conservation	**Ex-Situ Conservation**
On-site conservation	Off-site conservation
Conducted in natural habitats	Conducted outside natural habitats
More dynamic and sustainable	More controlled and static
Conserves entire ecosystem	Conserves selected species
Time-consuming but long-term	Used for urgent conservation

Q15

(a) Differentiate between National Park and Zoological Park

Basis	National Park	Zoological Park (Zoo)
Meaning	A protected natural area created to conserve wildlife, flora, fauna and ecosystems in their natural habitat.	A man-made facility where wild animals are kept in enclosures for public display, education and conservation.
Habitat	Natural or near-natural habitat.	Artificial or semi-natural enclosures.
Area	Covers a very large geographical area.	Covers a comparatively small and restricted area.
Human Interference	Human activities are strictly restricted.	Human management and intervention are very high.
Objective	Conservation of biodiversity and ecological processes.	Education, recreation, research and captive breeding.
Examples	Jim Corbett National Park, Kaziranga National Park.	Delhi Zoo, Alipore Zoo (Kolkata).

15(b) Differentiate between Biosphere Reserves and National Parks

Basis	Biosphere Reserves	National Parks
Meaning	Large protected areas aimed at conserving biodiversity along with sustainable use of resources.	Protected areas created mainly to conserve wildlife and ecosystems.
Legislation	Established under international (UNESCO) and national programmes.	Established by Central or State legislation.
Area Covered	Very large areas often including national parks and sanctuaries.	Smaller area compared to biosphere reserves.
Zonation	Divided into Core, Buffer and Transition zones.	No zonation; entire area is strictly protected.
Human Activities	Limited human activities allowed in buffer and transition zones.	Human activities like grazing, cultivation and forestry are prohibited.
Objective	Conservation + sustainable development + research.	Strict protection of flora and fauna.
Examples	Nilgiri Biosphere Reserve, Sundarbans Biosphere Reserve.	Ranthambore National Park, Gir National Park.

15(c) Differentiate between Grazing Food Chain and Detritus Food Chain

Basis	Grazing Food Chain	Detritus Food Chain

Starting Point	Begins with green plants (producers).	Begins with dead organic matter (detritus).
Energy Source	Solar energy captured by plants through photosynthesis.	Energy stored in dead remains of plants and animals.
Main Organisms	Herbivores → Carnivores.	Decomposers and detritivores like bacteria, fungi, earthworms.
Type of Organisms	Mostly large, visible organisms.	Mostly microorganisms and small organisms.
Ecosystem Type	Common in grasslands, forests and aquatic ecosystems.	Common in forest floor, soil and aquatic sediments.
Energy Flow	Less energy released back to environment.	Large amount of energy released through decomposition.

Q16

(a) Red Data Book

- The Red Data Book is an official document that records **rare, endangered and threatened species** of plants, animals and fungi.
- It is published by the **International Union for Conservation of Nature (IUCN)** and also by national agencies.
- It provides information on **population status, distribution and threats** to species.
- It helps in **conservation planning, legal protection and policy formulation**.
- Species listed in it receive **priority attention for conservation efforts**.

(b) Vermicomposting

- Vermicomposting is a method of composting using **earthworms** to decompose organic waste.
- Earthworms convert biodegradable waste into **nutrient-rich vermicompost**.
- It improves **soil fertility, structure and microbial activity**.
- It is an **eco-friendly, low-cost and sustainable** waste management technique.
- Commonly used earthworms include *Eisenia fetida* and *Eudrilus eugeniae*.

(c) Disappearing House Sparrows

- House sparrow populations have declined sharply in urban and rural areas.
- **Loss of nesting spaces** due to modern buildings is a major reason.
- **Excessive use of pesticides** reduces insects, the main food for chicks.

- **Electromagnetic radiation** from mobile towers affects navigation and breeding.
- Pollution and climate change have made cities **bird-unfriendly habitats**.

(d) Elephants as Endangered Species

- Asian and African elephants are listed as **Endangered** due to population decline.
- **Habitat loss and fragmentation** caused by deforestation and development are major threats.
- **Poaching for ivory tusks** fuels illegal wildlife trade.
- Human-elephant conflict leads to **retaliatory killings**.
- Conservation efforts include **Project Elephant, protected corridors and anti-poaching laws**.

(e) Extinction of Indian Tigers due to Loss of Genetic Diversity

- Habitat fragmentation has led to **small and isolated tiger populations**.
- Reduced **gene flow** causes inbreeding and loss of genetic diversity.
- Genetically weak populations are more **vulnerable to diseases and environmental changes**.
- Over 90% of tiger habitats have been lost due to human activities.
- Conservation strategies include **wildlife corridors and habitat restoration**.

(f) Jhum Cultivation

- Jhum cultivation is a **slash-and-burn shifting agriculture** practiced by tribal communities.
- Forest land is cleared, burned and cultivated temporarily.
- It leads to **deforestation, soil erosion and loss of biodiversity**.
- Short fallow periods reduce soil fertility.
- Though traditional and organic, it is **ecologically unsustainable** today.

(g) Jim Corbett National Park

- Jim Corbett National Park is the **oldest national park in India**, established in 1936.
- Located in **Uttarakhand**, it was originally named Hailey National Park.
- It was created to protect the **Bengal tiger** under Project Tiger.
- The park supports **rich biodiversity of flora and fauna**.
- It is also an important **eco-tourism and conservation centre**.

(h) Sacred Groves

- Sacred groves are patches of forest protected due to **religious and cultural beliefs**.

- Cutting trees and hunting are **strictly prohibited**.
- They conserve **rare, endemic and medicinal species**.
- Sacred groves help in **soil conservation, water recharge and climate regulation**.
- They act as **natural biodiversity hotspots**.

(i) Significance of Biodiversity Parks in Cities

- Biodiversity parks conserve **native flora and fauna in urban areas**.
- They improve **air quality and reduce urban heat island effect**.
- Provide habitats for **birds, insects and small animals**.
- Help in **environmental education and public awareness**.
- Act as **green lungs** and enhance the ecological health of cities.

Q.17. Project Tiger in India and its Importance for Human Survival

Project Tiger: An Overview

- **Project Tiger** is a wildlife conservation programme launched by the **Government of India in 1972**.
- The main objective of the project is **to protect the endangered Bengal tiger** and ensure its long-term survival.
- During the **1950s and 1960s**, tigers were hunted extensively for:
    - Sport
    - Their skin
    - Trophy collection
- Due to excessive hunting and habitat destruction:
    - Tiger population declined sharply from around **40,000 to about 1,800**.
- This alarming decline led to the launch of **Project Tiger**.

Key Features of Project Tiger

- India is home to nearly **60% of the world's wild tiger population**.
- Initially:
    - **9 Tiger Reserves** were selected
    - Covering an area of **16,339 sq. km**
- **Jim Corbett National Park** was the **first reserve** included under Project Tiger.
- At present:
    - **27 Tiger Reserves** are included

    - Covering around **37,761 sq. km**
- Tiger population increased from **268 tigers initially** to about **1,498 tigers** (as per later estimates).

Objectives of Project Tiger

- To ensure a **viable and stable population of tigers** in India.
- To provide **safe and secure habitats** for tigers and their prey species.
- To address major threats such as:
    - Habitat destruction
    - Loss of prey base
    - Poaching
    - Human–animal conflict
    - Competition with livestock and local communities
- To preserve areas of **biological importance** as natural heritage.
- To promote:
    - Wildlife management
    - Protection measures
    - Eco-development programmes for local communities.

Major Tiger Reserves in India

- Corbett National Park
- Ranthambore
- Sariska
- Periyar
- Sundarbans

Why Saving Tigers is Crucial for Human Survival

- Tiger is a **top predator** and a **keystone species**.
- It plays a vital role in maintaining the **ecological balance** of forests.
- Conservation of tigers helps in:
    - Protecting entire forest ecosystems
    - Maintaining food chains and biodiversity
- Healthy forests provide:

    - Clean air
    - Clean water
    - Climate regulation
    - Soil conservation
- Since humans depend on forests for survival:
    - Saving tigers indirectly means **saving human life-support systems**.
- Tigers are a symbol of:
    - Wilderness
    - Ecological health
    - National heritage
- Protecting tigers ensures conservation of:
    - Flora
    - Fauna
    - Genetic diversity

Conclusion

- Project Tiger is one of the **most successful wildlife conservation programmes** in India.
- Though poaching and habitat loss still exist, continuous efforts are being made.
- Saving tigers is not only about protecting a species but about **preserving ecosystems essential for human survival**.
- Therefore, **saving tigers ultimately means saving mankind itself**.

Q18. **Elaborate the following**

(a) Project Elephant

- **Project Elephant** was launched in **1992** by the **Government of India**, Ministry of Environment and Forests.
- It is a **Centrally Sponsored Scheme** aimed at conserving **wild Asian elephants**.
- The project provides **financial and technical assistance** to states.

Objectives of Project Elephant

- Protection of **elephants, their habitats and migration corridors**.
- Mitigation of **human–elephant conflict**.
- Research on **elephant ecology and management**.
- Welfare and proper care of **captive elephants**.

Implementation

- Implemented in **16 States/UTs**, including:
    - Assam, Kerala, Karnataka, Tamil Nadu, Odisha, Uttarakhand, West Bengal, etc.
- Around **32 Elephant Reserves** have been notified.
- **Singhbhum Elephant Reserve (Jharkhand)** was the first reserve.

Major Activities

- Prevention of **poaching and illegal ivory trade**.
- Scientific management of elephant habitats.
- Restoration of **degraded forests and corridors**.
- Veterinary care and training of forest staff.
- Census and monitoring of elephant populations.

MIKE Programme

- **Monitoring of Illegal Killing of Elephants (MIKE)** launched under **CITES (2003)**.
- Helps elephant-range countries in **management and law enforcement**.

(b) Vulture Breeding Programme

- Vultures play a **crucial ecological role** by disposing of animal carcasses.
- In the late **1990s**, a drastic decline in vulture population was observed in **India, Pakistan and Nepal**.
- The major cause identified was the **veterinary drug diclofenac**, which is fatal to vultures.

Government Initiatives

- **Vulture Recovery Plan (2004)** launched.
- Veterinary use of **diclofenac banned in 2006**.
- Promotion of **safe alternative drugs**.

Role of BNHS

- **Bombay Natural History Society (BNHS)** led conservation efforts.
- Focus areas:
    - Conservation breeding
    - Research and monitoring
    - Public awareness
    - Release of captive-bred vultures into the wild

Vulture Conservation Breeding Centres

- First centre established at **Pinjore, Haryana (2005)**.
- Other centres located at:
  - Rani (Assam)
  - Buxa (West Bengal)
  - Bhopal (Madhya Pradesh)
- Joint initiative of **Central Zoo Authority (CZA)** and BNHS.
- Focus on conserving:
  - White-backed Vulture
  - Long-billed Vulture
  - Slender-billed Vulture

(c) Project Great Indian Bustard

- The **Great Indian Bustard (GIB)** is one of the **heaviest flying birds**, weighing up to **15 kg**.
- It is a **flagship species of grassland ecosystems**.
- Declared **Critically Endangered** by **IUCN in 2011**.
- Less than **200 birds** remain, with about **100 in Rajasthan**.

Project Launch

- **Project Great Indian Bustard** launched by **Rajasthan Government** on **World Environment Day (2018)**.
- Budget allocation of **₹12 crore**.

Objectives

- Recovery of GIB population.
- Protection of **grassland habitats**.
- Prevention of extinction similar to the **Indian Cheetah**.
- Awareness and involvement of local communities.

Significance

- Indicates the **health of grassland ecosystems**.
- Loss of GIB reflects severe degradation of grasslands.

(d) Crocodile Conservation Project

- **Project Crocodile** launched in **1975**.
- Supported by:
    - **United Nations Development Programme (UNDP)**
    - **Food and Agriculture Organization (FAO)**.
- Considered one of the **most successful conservation projects** in India.

Objectives

- Protection of crocodilians in their **natural habitats**.
- Restoration of depleted populations.
- Promotion of **scientific research and management**.

Major Conservation Methods

- Creation of crocodile sanctuaries.
- **'Grow and Release' technique**:
    - Collection of eggs from natural nests
    - Artificial incubation
    - Rearing hatchlings in captivity
    - Marking and releasing into protected areas
    - Monitoring survival after release

Research Areas

- Habitat studies
- Behavioural biology
- Feeding, growth and reproduction
- Population structure and census

Community Participation

- Involvement of local people.
- Protection of fishermen's livelihoods.
- Promotion of crocodile farming as an alternative income source.

(e) Save Western Ghats Movement (SWGM)

- **Save Western Ghats Movement** was a landmark **environmental mass movement** in India.
- Based on the principle of:

- Environmental protection
- People's participation
- Sustainable livelihoods

Background

- Originated due to **deforestation and ecological degradation** in Western Ghats.
- In **October 1986**, a national consultation led to the decision to organize a mass march.

The March

- A **100-day march** covering the entire Western Ghats.
- Two teams started from:
  - **Navapur (North)**
  - **Kanyakumari (South)**
- Over **160 organizations** and thousands of individuals participated.
- More than **60 public meetings** conducted.

Impact

- Generated massive public awareness.
- Strong youth participation.
- Extensive national and international media coverage.
- Highlighted the need for an **integrated ecological and social approach**.

Limitations

- Lack of consensus on follow-up action plans.
- Could not fully convert enthusiasm into long-term sustainable development policies.

UNIT-VII

HUMAN COMMUNITIES AND THE ENVIRONMENT

Q.1 Population Explosion: Meaning and Its Impact on Environment, Human Health and Welfare

Meaning of Population Explosion

Population explosion refers to the **rapid and uncontrolled increase in human population** over a short period of time.
In the last hundred years, the world population has increased dramatically due to:

- High birth rates
- Decline in death rates
- Improvement in medical facilities
- Increase in life expectancy

According to estimates, the world population was about **7.2 billion in 2013** and is expected to reach **around 9.6 billion by 2050**.

Causes of Population Explosion

1. **High birth rate** compared to death rate
2. **Population momentum** – large young population entering reproductive age
3. **Improved healthcare and sanitation**
4. **Lack of awareness about family planning**
5. **Poverty and illiteracy** in developing countries

Uneven Distribution of Population

- World population is unevenly distributed among **233 countries**
- A major portion of the population lives in **developing countries**
- These countries face maximum pressure on natural resources

Impact of Population Explosion

1. Impact on Environment

Population explosion puts **excessive pressure on natural resources**, leading to environmental degradation.

Major Environmental Impacts:

- **Deforestation** for agriculture, housing and industries
- **Depletion of natural resources** such as water, fossil fuels and minerals

- **Loss of biodiversity** due to habitat destruction
- **Increase in pollution** (air, water, soil and noise)
- **Generation of huge amounts of waste**

Uneven utilization of resources and excessive waste generation have created an **interrelated web of environmental problems**.

Example (India):
India faces an acute **resource crisis** due to its large population. There is fierce competition for land, water, food and energy, affecting the quality of life.

2. Impact on Human Health

Population explosion directly affects human health because health is closely linked with **food, housing, sanitation and environment**.

Effects on Health:

1. **Food scarcity and malnutrition**
   - Insufficient food leads to poor health and low productivity
2. **Spread of diseases**
   - Overcrowding and poor sanitation increase communicable diseases
3. **Pollution-related diseases**
   - Air and water pollution cause respiratory, skin and water-borne diseases
4. **Poor living conditions**
   - Slums and congested housing increase health risks

3. Impact on Human Welfare

- Low per capita income
- Poverty and unemployment
- Poor education and illiteracy
- Reduced standard of living
- Slower economic development

Thus, population explosion negatively affects **economy, progress and welfare of nations**.

Q.2 Impact of Increasing Consumerism on Environment (Resource Depletion & Pollution)

Meaning of Consumerism

Consumerism refers to the **excessive consumption of goods and services**, often beyond basic needs, driven by modern lifestyles, advertising and easy availability of products.

Present Consumption Model

The current economic system follows a **linear model**:

1. Extraction of raw materials
2. Manufacturing and production
3. Distribution and consumption
4. Disposal as waste

This model is **unsustainable** in the long run.

Impact of Consumerism on Environment

1. Resource Depletion

- Non-renewable resources like **coal, oil and minerals** are being exhausted
- Over-extraction disturbs **natural cycles**, such as the water cycle
- Fossil fuel burning releases large amounts of greenhouse gases

2. Pollution

- Industrial production causes **air, water and soil pollution**
- Toxic chemicals are released during manufacturing
- Pollution affects ecosystems and human health

3. Case Study: Fast Food Chains (KFC, Pizza Hut, etc.)

- Intensive livestock farming leads to:
    - **Deforestation**
    - **Land degradation**
    - **Water contamination**
- For every **1 pound of meat produced**, about **5 pounds of topsoil** is lost
- **190 gallons of water** is required per animal per day
- Nearly **40% of global grain production** is used for animal farming
- In the USA, **70% of grain** is fed to livestock

This shows how consumer food habits contribute to environmental degradation.

4. Outsourcing of Polluting Industries

- Hazardous industries are shifted to **developing countries**
- Environmental laws are weakly enforced
- Results in health hazards for local populations

5. Planned Obsolescence

- Products are designed with **short life spans**
- Encourages frequent replacement
- Leads to excessive waste generation

6. Waste Generation

- Waste levels have **doubled in the last 30 years**
- Disposal of plastic, e-waste and toxic waste is a major challenge
- Incineration releases harmful gases

Negative Externalities

Environmental and social costs not included in product prices are called **negative externalities**, such as:

- Pollution
- Health impacts
- Exploitation of labour

Solutions: Towards a Sustainable Future

1. Circular Economy

- Repair, reuse and recycle products
- Reduce waste generation

2. Role of Government

- Enforce strict environmental laws
- Promote sustainable production

3. Role of Companies

- Eco-friendly product design
- Responsible manufacturing

4. Role of Consumers

- Informed and responsible purchasing
- Reduce over-consumption

Q. 3 What is a Disaster? Explain Commonly Occurring Disasters in India

(a) Disaster – Meaning

A **disaster** is a **sudden and catastrophic event** that causes **large-scale loss of life, property and environmental damage**.
The impact of a disaster depends on:

- Geographical location
- Climate conditions
- Type of hazard
- Vulnerability of people

Disasters disturb the **social, economic, political and cultural life** of affected areas.

Types of Disasters

1. Natural Disasters

Natural disasters occur due to **natural forces** and result in human, economic and environmental losses.

Examples:

- Earthquakes
- Floods
- Cyclones
- Landslides
- Droughts
- Volcanic eruptions

Example: **2015 Nepal EarthquakE**

2. Man-Made Disasters

These disasters are caused due to **human activities**.

Examples:

- Industrial accidents
- Nuclear explosions
- Chemical leakage
- War and riots
- Environmental pollution

Example: **1984 Bhopal Gas Tragedy**

Vulnerability and Disaster

Disasters occur when **hazards** interact with **vulnerable conditions**, such as:

- Poverty
- Illiteracy

- Poor infrastructure
- Unsafe housing
- Environmental degradation
- Rapid urbanization

(b) Flood

A **flood** is an overflow of water beyond its normal limits, submerging land that is usually dry.

Causes of Floods

- Heavy rainfall
- Overflowing rivers
- Cloudbursts
- Melting of snow
- Cyclones
- Dam failure

Impacts of Floods

- Loss of human life and livestock
- Destruction of crops and houses
- Soil erosion
- Spread of water-borne diseases
- Homelessness and famine-like conditions

Floods are both **natural and man-made disasters**.

(c) Earthquake

An **earthquake** is the shaking or trembling of the earth's crust caused by a sudden release of energy.

Measurement

- Measured using a **seismograph**

Causes of Earthquakes

1. **Tectonic Movements**
    - Sudden movement of crustal plates
    - Most common cause
      Example: Nepal Earthquake (2015)
2. **Volcanic Eruptions**

    - Occur during explosive volcanic activity

Impacts of Earthquakes

- Collapse of buildings, bridges and dams
- Damage to transport and power systems
- Landslides and floods
- Loss of life and property
- Environmental degradation

(d) Cyclone

A **cyclone** is a violent storm with **high-speed winds**, heavy rainfall and storm surges.

Different Names of Cyclones

- **Bay of Bengal:** Depressions / Cyclones
- **Caribbean Sea:** Hurricanes
- **China Sea:** Typhoons
- **Australia:** Willy-willy
- **USA & West Africa:** Tornadoes

Damages Caused

- Destruction of houses
- Flooding
- Loss of crops
- Power and communication failure

(e) Landslide

A **landslide** is the downward movement of rock, soil or debris along a slope due to gravity.

Causes of Landslides

- Heavy rainfall
- Earthquakes
- River and coastal erosion
- Overgrazing
- Deforestation
- Construction and mining

Common Areas

- Mountainous regions
- River banks
- Coastal areas

Q. 4 Disaster Management and Its Tips

Meaning of Disaster Management

Disaster Management is a **systematic process of planning, organizing and managing resources** to:

- Prevent disasters
- Reduce their impact
- Respond effectively
- Recover quickly

Disaster Management Cycle

1. Mitigation

Actions taken to **reduce the severity** of disasters.

- Can be done **before, during or after** disasters

2. Preparedness

Measures taken to **prepare people and systems** to face disasters.

- Training
- Early warning systems
- Emergency planning

3. Response

Immediate actions taken **during and after a disaster**.

- Rescue
- Relief
- Medical aid

4. Recovery

Includes:

- Rehabilitation
- Reconstruction
- Restoration of services

Disaster Management Tips

1. Earthquake

During Earthquake

- Take cover under strong furniture
- Stay close to the floor
- Move away from windows and heavy objects
- Never use lifts
- If outdoors, move to open areas

After Earthquake

- Give first aid to injured persons
- Wear shoes to avoid glass injuries
- Check building safety
- Use stairs during evacuation
- Be alert for aftershocks
- Stay away from beaches (tsunami risk)

2. Floods

Before Floods

- Identify safe shelters and routes
- Prepare emergency kits
- Stay informed through radio/TV/mobile alerts
- Move to higher ground if warned

During Floods

- Drink boiled water
- Avoid floodwater
- Do not drive through flooded roads
- Avoid standing water with electrical wires

3. Landslides

- Stay alert during heavy rains
- Listen for unusual sounds
- Watch changes in water flow

- Evacuate immediately if danger is suspected
- Avoid landslide-prone areas
- Help injured after the slide

4. Cyclones

- Stock food, water and essential supplies
- Move to cyclone shelters
- Ensure medicines and torches
- Use generators during power failure
- Maintain communication systems

Q. 8 Reasons for Floods in Jammu & Kashmir (2015)[2018 May]

The floods in **Jammu and Kashmir (2015)** were caused due to a combination of **natural and human-made factors**.

Major Reasons

1. **Unprecedented Heavy Rainfall**
    - Continuous and intense rainfall over a short period
    - Led to **cloudbursts**
    - Rivers **Jhelum, Chenab and Tawi** crossed danger levels
2. **Overflow of Rivers and Tributaries**
    - Flooding of low-lying catchment areas for more than **two weeks**
3. **Poor Drainage Capacity**
    - Rivers and channels lacked sufficient carrying capacity
4. **Unplanned Urbanisation**
    - Construction over floodplains
    - Encroachment on lakes, ponds and wetlands
5. **Encroachment of Water Bodies**
    - Out of **1230 lakes and water bodies**, over **50% encroached** in the last 100 years
    - Natural drainage network collapsed
6. **Deforestation in Jhelum Basin**
    - Excessive siltation in rivers and lakes
    - Reduced water-holding capacity

Conclusion

The disaster was intensified due to **ecological degradation, deforestation and poor urban planning** along with extreme rainfall.

Q. 9 Socio-Economic Impacts of Increasing Floods in India & Disaster Management

[2018 Nov]

Socio-Economic Impacts of Floods

1. Loss of Life and Property

- Deaths due to drowning and flash floods
- Damage to houses, roads, bridges and crops
- Spread of water-borne diseases

2. Loss of Livelihoods

- Agriculture and industry severely affected
- Disruption of power, transport and communication
- Long-term unemployment

3. Reduced Purchasing & Production Power

- Loss of income and land value
- Increased vulnerability of flood-plain communities
- High rehabilitation costs divert development funds

4. Mass Migration

- Displacement of people
- Migration to cities → slums & overcrowding
- Social and economic stress

5. Psychosocial Effects

- Trauma, stress and depression
- Loss of family members and homes
- Long-term psychological impact, especially on children

6. Hindrance to Economic Development

- High cost of relief and reconstruction
- Discourages investment
- Inflation and slowed regional growth

7. Political Implications

- Public dissatisfaction with government
- Social unrest and loss of trust in authorities

Disaster Management in Flood-Affected Areas

Before Floods

- Identify safe shelters and evacuation routes
- Prepare emergency kits
- Monitor weather alerts
- Move to higher ground when warned

During Floods

- Drink boiled water
- Avoid flooded areas and electric lines
- Do not drive through floodwaters
- Disinfect surroundings

Q. 10 Role of Disaster Management in Controlling Recurrent Floods in Indian Plains

[2017 May]

India is highly prone to **recurrent floods**, especially in plains.

Role of Disaster Management

Before Floods

- Flood forecasting and early warning systems
- Strengthening embankments and drainage
- Emergency preparedness kits
- Public awareness and evacuation planning

During Floods

- Safe drinking water
- Medical aid and relief camps
- Avoid standing water and damaged roads

Outcome

Proper disaster management **reduces loss of life, property and economic damage**.

Q. 11 Impact of Dams on Livelihood – Tehri Dam Case Study

Background

- Built on **Bhagirathi and Bhilangana rivers (Uttarakhand)**
- One of the world's highest dams

Negative Impacts

1. **Seismic Vulnerability**
    - Located in earthquake-prone zone
2. **Displacement**
    - Submergence of **5200 hectares**
    - 1 town, 37 villages fully & 88 villages partially affected
3. **Loss of Livelihood**
    - Farmers and tribals lost land
    - Cultural and religious sentiments hurt

Rehabilitation Issues

- Cash compensation instead of land
- Families displaced individually, not as communities
- Poor awareness of rights
- Urban land pressure in resettlement areas

Q. 12 Short Notes

(a) Carbon Footprint

Meaning

- Carbon footprint refers to the **total amount of greenhouse gases**, mainly **carbon dioxide (CO_2)**, released into the atmosphere due to **human activities**.

Sources of Carbon Footprint

- Use of vehicles running on petrol and diesel
- Industrial and factory emissions
- Electricity generation using coal and thermal power
- Deforestation
- Excessive use of plastic and fossil fuels

Effects

- Causes **global warming**

- Leads to **climate change**
- Results in **rise in sea level**
- Causes **extreme weather events** like floods and droughts

Reduction Measures

- Use of renewable energy such as solar and wind power
- Using public transport and car pooling
- Use of energy-efficient appliances
- Tree plantation
- Recycling and reduction of waste

Conclusion

- Reducing carbon footprint is very important for **environmental protection and sustainable development**.

(b) Environmental Movements

Meaning

- Environmental movements are **collective efforts** by **local people, NGOs and activists** to protect the environment.

Objectives

- Conservation of forests, rivers and wildlife
- Protection of the livelihood of local communities
- Opposition to environmentally harmful development projects
- Promotion of sustainable development

Characteristics

- Active participation of people
- Use of non-violent methods
- Focus on social and environmental justice
- Mostly grassroots-level movements

Examples

- Chipko Movement
- Narmada Bachao Andolan
- Silent Valley Movement

Importance

- Environmental movements forced governments to adopt **environment-friendly policies**.

(c) Chipko Movement

Background

- Started in **1973**
- Location: **Garhwal region, Uttarakhand**
- Leader: **Sunderlal Bahuguna**

Meaning

- The word "Chipko" means **to hug**.
- Villagers hugged trees to prevent them from being cut.

Main Features

- Protest against commercial deforestation
- Major participation of women
- Non-violent method of protest

Impact

- Government imposed restrictions on tree cutting
- Increased awareness about forest conservation
- Encouraged community participation

Significance

- The Chipko Movement proved that **people's participation is essential for environmental protection**.

(d) Silent Valley Movement

Background

- Started in **1973**
- Location: **Kerala**
- Issue: Proposed hydroelectric power project

Objective

- To protect **tropical evergreen forests** and **rare species** like the lion-tailed macaque.

Movement Highlights

- Involvement of scientists, environmentalists and local people
- Long-term protests and awareness campaigns

Outcome

- Hydroelectric project was cancelled
- Silent Valley was declared a **National Park in 1985**

Importance

- Highlighted the importance of **biodiversity conservation**.

(e) Bishnois of Rajasthan

Introduction

- Bishnois are a **nature-loving religious community** of Rajasthan.

Beliefs

- Protection of trees and animals is a religious duty
- Hunting of wildlife is strictly prohibited
- Environmental conservation is considered a spiritual practice

Historical Example

- **Khejarli Massacre (1730)**
- 363 Bishnois sacrificed their lives to protect Khejri trees

Importance

- Bishnois are regarded as **India's earliest environmental protectors**.

(f) National Green Tribunal (NGT)

Establishment

- Established in **2010**
- Formed under the **National Green Tribunal Act**

Objectives

- Speedy disposal of environmental cases
- Protection and conservation of the environment
- Conservation of forests and wildlife

Powers

- Can impose penalties on polluters
- Can stop environmentally harmful projects
- Applies principles such as:
    - Polluter Pays Principle

- Sustainable Development

Importance

- NGT has made environmental justice **fast, effective and accessible**.

(g) Green Auditing

Meaning

- Green auditing refers to the **assessment of an organization's environmental performance**.

Objectives

- Compliance with environmental laws
- Control of pollution
- Efficient use of natural resources

Benefits

- Reduction in environmental risks
- Improvement in public image
- Long-term cost savings

Conclusion

- Green auditing helps organizations become **eco-friendly and environmentally responsible**.

(h) Green Energy

Meaning

- Green energy is energy obtained from **renewable and environment-friendly sources**.

Sources

- Solar energy
- Wind energy
- Hydropower
- Biomass energy

Advantages

- Causes minimal or no pollution
- Renewable and unlimited sources
- Reduces carbon emissions
- Supports sustainable development

Importance

- Green energy helps in reducing the impact of **climate change**.

(i) Environmental Ethics

Meaning

- Environmental ethics refers to the **moral responsibility of humans towards nature**.

Key Principles

- Respect for nature
- Conservation of natural resources
- Protection of future generations
- Sustainable use of the environment

Importance

- Prevents environmental degradation
- Promotes harmony between humans and nature
- Encourages an eco-friendly lifestyle

Conclusion

- Environmental ethics ensure that **development takes place without harming the environment**.

Q. 13 Difference: Chipko Movement vs Beej Bachao Andolan

Chipko Movement	Beej Bachao Andolan
Started in 1970s	Started in 1990s
Forest conservation	Seed conservation
Hugging trees	Saving indigenous seeds
Led by Sunderlal Bahuguna	Led by Vijay Jardhari
Focus on trees	Focus on agro-biodiversity

Q. 14 Environmental Philosophy

Environmental philosophy studies **human–nature relationships**.
It examines:

- Value of nature
- Ethical responsibility
- Sustainable living

- Justice to future generations

Core Principles

- Justice & sustainability
- Sufficiency & compassion
- Participation & solidarity

It guides solutions for **pollution, climate change and ecological crises**.

Q.15. Explain the role of Indian religions and culture in environmental conservation

Introduction

- All Indian religions and cultures promote **respect for nature**.
- Protection of the environment is considered a **moral and spiritual duty**.
- Nature is seen as sacred and divine.

Role of Indian Religions

1. Hinduism

- Nature and God are considered **one and inseparable**.
- Rivers, mountains, trees, animals and land are worshipped.
- Belief in **Ahimsa (non-violence)** discourages killing of animals.
- Concept of rebirth teaches respect for all life forms.
- Sacred rivers like **Ganga**, trees like **Peepal, Banyan and Neem** are protected.

2. Christianity

- Bible emphasizes **human responsibility to protect nature**.
- God is the owner of the natural world; humans are caretakers.
- The concept of "dress and keep the Garden of Eden" means **manage and protect nature**.

3. Islam

- Quran teaches **harmony, balance and unity** in nature.
- Humans are guardians of nature, not exploiters.
- Destruction of nature leads to **accountability after death**.
- Natural laws must be respected.

4. Sikhism

- Guru Nanak regarded nature as **divine**.
- Humans are an integral part of nature.

- Balance between elements of nature is necessary for survival.
- Guru Granth Sahib emphasizes creation and dissolution under divine command.

5. Jainism

- Core principle is **non-violence towards all living beings**.
- Teaches minimal use of resources.
- Promotes environmental harmony through spirituality.

6. Buddhism

- Advocates compassion, tolerance and non-injury.
- Belief in karma teaches that harming nature leads to suffering.
- Encourages simple living and respect for all life forms.

Role of Indian Culture

- Nature is worshipped as **Mother Earth (Dharti Maa)**.
- Rivers like Ganga are revered as **Ganga Maa**.
- Trees, animals and mountains are considered sacred.
- Sacred groves are protected from exploitation.
- Tribal communitie act as protectors of forests and biodiversity.

Conclusion

- Indian religions and culture strongly support **environmental conservation and sustainable living**.

Q.16. "Cultural practices are powerful tools to protect the environment but can also be threats." Justify

Introduction

- Culture shapes how humans interact with nature.
- Cultural practices can both **protect** and **harm** the environment.

Cultural Practices Protecting Environment

1. Chipko Movement

- Started in Uttarakhand.
- People hugged trees to prevent deforestation.
- Promoted forest conservation and ecological balance.

2. Sacred Groves

- Forest areas protected due to religious beliefs.

- Cutting trees or hunting is prohibited.
- Help in conservation of biodiversity.

3. Animal Protection Traditions

- Certain animals are worshipped and protected.
- Prevents extinction of species.

Cultural Practices Threatening Environment

- Excessive animal protection may lead to **man-animal conflict**.
- Habitat loss forces animals into human settlements.
- Results in damage to crops, property and human life.
- Over-emphasis on certain traditions may ignore ecological balance.

Conclusion

- Culture plays an important role in conservation.
- However, cultural practices must be **scientifically balanced and context-based**.

Q.17. Write a short note on Environmental Education

Meaning

- Environmental education is the process of **creating awareness about environmental issues**.
- It helps people understand the difference between development and sustainable development.

Importance

- Inculcates environmental values from an early age.
- Encourages responsible behaviour towards nature.
- Develops problem-solving skills related to environmental issues.

Implementation

- Made a compulsory subject in **2004**.
- Included in school and university curriculum.
- Supported by government and NGOs.

Awareness Programmes

- Clean Environment Programme
- Plant a Tree Campaign
- Plastic-Free Day

Important Environmental Days

- World Environment Day – **5 June**
- Ozone Day – **16 September**
- Wetland Day – **2 February**
- Water Day – **22 March**
- Desertification Day – **17 June**

Conclusion

- Environmental education is essential for **sustainable future and conservation efforts**.

Q.18. What are the objectives of Environmental Education?

Objectives

1. Knowledge

- To provide understanding of environment and ecosystems.

2. Awareness

- To create awareness about pollution and environmental degradation.

3. Attitudes

- To develop positive values towards environmental protection.

4. Skills

- To develop skills for solving environmental problems.

5. Participation

- To encourage active involvement in environmental decision-making.

6. Evaluation

- To assess environmental policies and actions critically.

Conclusion

- Environmental education develops **informed, skilled and responsible citizens**.

Q.19. Role of Environmental Communication and Public Awareness in Environmental Protection

Introduction

- Many environmental problems arise due to **lack of awareness**.
- Environmental communication helps spread knowledge and responsibility.

Major Measures

1. Environmental Education in Schools

- Integrating environmental studies into curriculum.
- Builds awareness from a young age.

2. Role of Media

- Newspapers, radio, television and internet spread awareness.
- Highlights environmental issues and solutions.

3. Use of Maps, Radio and Videos

- Participatory mapping creates community involvement.
- Radio programmes educate rural population.
- Videos and campaigns influence public behaviour.

Public Participation

- Encourages environmental citizenship.
- Helps people understand their rights and duties.
- Strengthens enforcement of environmental laws.

Conclusion

- Public awareness and communication are essential for **environmental protection and sustainable development**.

Q.20. Contribution of Women in Environmental Protection in India

Introduction

- Women play a crucial role in environmental conservation.
- Their close relationship with natural resources gives them a holistic perspective.

Major Women Environmentalists

1. Gaura Devi

- Leader of women in Chipko Movement (1974).
- Hugged trees to stop deforestation.
- Head of Mahila Mangal Dal.

2. Medha Patkar

- Leader of Narmada Bachao Andolan.
- Opposed displacement due to large dams.
- Used peaceful protests like fasting.

3. Sunita Narain

- Director General of Centre for Science and Environment.
- Editor of *Down to Earth* magazine.
- Chaired Tiger Task Force in 2005

4. Vandana Shiva

- Eco-feminist and environmentalist.
- Founder of Navdanya Movement.
- Conserved thousands of seed varieties.
- Awarded Right Livelihood Award (1993).

5. Maneka Gandhi

- Animal rights activist.
- Founder of People for Animals (1994).
- Promoted animal welfare and non-violence.

Q.21. Write a note on the introduction of CNG in Delhi

Introduction

- Rapid increase in vehicles since the 1980s led to **severe air pollution in Delhi**.
- According to Delhi Traffic Police, around **80,000 vehicles pass through Delhi daily at night**, adding to pollution.
- Both private and public transport expanded due to population growth.

Role of the Supreme Court

- The **Supreme Court of India** intervened to reduce vehicular pollution.
- It promoted the use of **alternative and cleaner fuels**.
- In **1998**, it ordered the conversion of all **commercial vehicles** to CNG.

Introduction of CNG

- **Compressed Natural Gas (CNG)** was introduced as a cleaner fuel compared to petrol and diesel.
- Use of **CNG and LPG** was made mandatory for:
    - Buses
    - Taxis
    - Auto-rickshaws
    - Other commercial vehicles

- Delhi developed one of the **largest CNG vehicle fleets in the world**.

Advantages of CNG

- Produces **less air pollution**
- Reduces emission of harmful gases
- Environment-friendly fuel
- Helps in improving air quality

Problems Faced

1. **Insufficient CNG stations**, leading to long queues.
2. **Safety issues** due to improper installation of CNG kits.
3. **Shortage of trained technicians** for maintenance and refuelling.

Conclusion

- Despite initial challenges, the introduction of CNG has **significantly reduced air pollution in Delhi**.
- It is considered a **successful step towards cleaner urban transport**.

Q.22. What are the objectives of 'Swachh Bharat Abhiyan'? What are the challenges and how can it be made more effective?

Introduction

- **Swachh Bharat Abhiyan** is a nationwide cleanliness campaign launched by the Government of India.
- It was launched on **2nd October 2014** by Prime Minister **Narendra Modi**.
- The mission aims to clean **streets, roads and public infrastructure** in urban and rural areas.

Objectives of Swachh Bharat Abhiyan

1. **Elimination of Open Defecation**
    - Construction of household and public toilets.
    - Promote safe sanitation practices.
2. **Improvement in Quality of Life**
    - Clean surroundings lead to better health and hygiene.
3. **Awareness and Behaviour Change**
    - Educate people about sanitation and cleanliness.
    - Encourage hygienic habits.

4. **Promotion of Sustainable Sanitation**
    - Encourage eco-friendly and cost-effective sanitation technologies.
5. **Solid and Liquid Waste Management**
    - Proper collection, segregation and disposal of waste.

Challenges of Swachh Bharat Abhiyan

1. **Lack of Awareness**
    - Many people do not recognize sanitation as a serious issue.
2. **High Illiteracy Rate**
    - Makes it difficult to communicate health and hygiene messages.
3. **Negative Public Attitude**
    - "Someone else will clean" mentality.
    - Lack of personal responsibility.
4. **Poor Maintenance**
    - Toilets constructed but not properly maintained.

Phases of Implementation

1. **Planning Phase**
    - Identification of problem areas.
    - Preparation of action plans.
2. **Implementation Phase**
    - Construction of toilets.
    - Waste management activities.
3. **Sustainability Phase**
    - Ensuring long-term usage and maintenance.
    - Continuous monitoring and follow-up.

Ways to Make the Movement More Effective

- Increase **public awareness campaigns**.
- Focus on **behavioural change**, not just infrastructure.
- Involve **schools, anganwadis and local communities**.
- Encourage **people's participation and accountability**.
- Ensure proper **maintenance of sanitation facilities**.

Conclusion

- Swachh Bharat Abhiyan is a landmark initiative for a **clean and healthy India**.
- Its success depends on **active participation, awareness and long-term sustainability**.

Q.23. Briefly describe the following

(a) National Environment Awareness Campaign (NEAC)

- The **National Environment Awareness Campaign (NEAC)** is supported by the **Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India**.
- The main objective of NEAC is to **create environmental awareness** among people.
- Special emphasis is given to **biodiversity conservation**, climate change, waste management and sustainable practices.
- Under this programme, **financial assistance** is provided to:
    - Non-Governmental Organisations (NGOs)
    - Educational institutions
    - Women's organisations
    - Youth organisations
- Around **12,000 organisations** participate in awareness and action-based programmes related to environmental protection.
- **Aranya**, an environmental NGO, plays an important role under NEAC by:
    - Creating awareness among **farmers and tribal communities**
    - Encouraging adoption of **biodiversity-friendly practices**
- NEAC campaigns are conducted **annually since 2007**.
- Activities include:
    - Tree plantation
    - Climate change awareness
    - Promotion of indigenous seed varieties
- Every year, NEAC adopts a **specific theme**, such as:
    - Solid Waste Management
    - Biodiversity Conservation
    - Climate Change
    - Forests for Sustainable Livelihood
    - Combating Desertification, Land Degradation and Drought

(b) National Green Corps (NGC)

- **National Green Corps (NGC)** is a programme of the **Ministry of Environment and Forests, Government of India**.
- It operates through **School Eco-Clubs** across the country.
- About **1,20,000 schools** are covered under this programme.
- Each NGC Eco-Club consists of **30–50 students**, known as **NGC Cadets**.
- The main objective is to **develop environment-friendly attitudes and actions among students**.
- Activities undertaken by NGC students include:
    - Biodiversity conservation
    - Water conservation
    - Energy conservation
    - Waste management
    - Land use planning and resource management
- Special focus is given to **local environmental issues**.
- Popular activities include:
    - Rainwater harvesting
    - Tree plantation
    - Composting biodegradable waste
- Each district has about **250 Eco-Clubs**.
- Financial support provided:
    - ₹2,500 per Eco-Club annually
    - ₹25,000 per district for monitoring
- Each state has a **State Nodal Officer** to implement the programme.
- States like **Telangana, Andhra Pradesh, Karnataka, Gujarat, Kerala, Maharashtra, Delhi, Himachal Pradesh** etc. show strong participation.
- NGC is considered the **largest school-based environmental programme in the world**.
- In some states, NGC cadets are trained as a **Green Brigade** and even wear uniforms during national celebrations.

(c) Eco Club Programme

- The **Eco Club Programme** aims to create **environmental awareness among students**.

- It encourages students to **actively participate in protecting the environment**.
- The programme starts in classrooms and gradually spreads to:
    - Schools
    - Communities
    - Society at large
- Eco Clubs help students understand **environmental management policies**.
- Schools participating successfully are awarded a **Green Flag**, which is a mark of environmental excellence.
- In Delhi, around **2000 Eco-Clubs** are functioning in:
    - Government schools
    - Aided schools
    - Private schools
    - Colleges
- The **Department of Environment** provides a grant of **₹20,000 per Eco-Club**.
- Major activities include:
    - Tree plantation and greenery drives
    - Water conservation awareness
    - Waste reduction and segregation
    - Awareness against burning of waste
    - Reducing use of plastic bags
    - Organising quizzes, rallies, poster making, essay competitions and street plays
    - Nature trails in forests, parks and wildlife sanctuaries
- The Eco-Schools Programme follows:
    - **Seven Steps Framework**
    - Eco-Schools themes
    - Assessment for Green Flag certification
- Teachers are trained as **Master Trainers**.
- Students are encouraged to participate in **field activities** to understand biodiversity and conservation.
- The programme helps in creating **environmentally responsible future citizens**.

PYQS

Ques1) Expand the following:

1. **UNEP-** United nations environment programme
2. **CITES-** Convention on international trade in endangered species of wild fauna and flora +
3. **NGC-** National Green Corps
4. **CBD-** Convention on Biological Diversity +
5. **WPA-** Wildlife Protection Act
6. **IUCN-** International Union for conservation of nature+
7. **NGT-** National Green Tribunal
8. **EPA-** Environment protection act +
9. **UNFCCC-**United nations framework convention on climate change +
10. **MAB-** Man and the biosphere programmes
11. **MOEFCC-** Ministry of environment, forest and climate change
12. **OPCW-** Organisation for the prohibition of chemical weapons
13. **NMEEE-** National mission for enhanced energy efficiency
14. **IPCC-** Intergovernmental panel on climate change
15. **BOD-** Biological oxygen demand
16. **GPP**- Gross primary productivity
17. **ENSO-** EL Nino- southern oscillation
18. **UNCED-** United nations conference on environment and development
19. **INC-** Intergovernmental Negotiating 9Committee
20. **CWC-** Chemical weapons convention
21. **TACC-** Territorial approach to climate change
22. **NAPCC-** National action plan on climate change
23. **IGCC-** Integrated gasification combined cycle
24. **IPM-** Integrated pest management
25. **VCBC-** Vulture Conservation breeding centre
26. **VCC-** Vulture care centre
27. **SWGM-** Save western ghats movement

Ques2) Write short notes on the following (2.5 * 4=10)

1. **Endemic species**
2. **Biological invasions**
3. **Swachh bharat abhiyan**
4. **Biodiversity**
5. **Global warming-**
6. **Endangered species-**
7. **Acid rain**
8. **Carbon footprints**
9. **Montreal protocol**
10. **Sacred groves**

Ques3) Difference between:

1. **National green tribunal V/S Central pollution control board**
2. **Biosphere reserves V/S Biodiversity Hotspot**
3. **Resettlement V/S Rehabilitation**
4. **In- situ conservation V/S ex-situ conservation +**
5. **Climate V/S weather**
6. **Wildlife protection act V/S Forest conservation act**

Ques4) State whether true/ false.

1. **Noise pollution can increase stress and anxiety levels-** True
2. **Carbon monoxide is not greenhouse gas contributing to global warming.**- True
3. **Greenpeace was founded in 19711 to support environmental causes and promote non-violent test-**True.
4. **Ozone holes are more pronounced at the equator –** False, Ozone holes are most severe over the polar regions, especially Antarctica, not equator.
5. **Project tiger was launched on April 1, 1983 by the government of India. –** False, Project tiger was actually launched on April 1, 1973 to protect Royal Bengal Tigers in India.

Ques5) Fill in the blanks.

1. Western ghats **and** Eastern Himalayas **are two biodiversity hotspots in India.**
2. Conservation in the natural habitat are called In-Situ conservation**.**
3. Ozone layer in Stratosphere **protects us from UV radiations.**
4. Brundtland commission published their report called Our Common Future **in 1987.**
5. International union for conservation of nature and natural resources (IUCN) **is the organisation responsible for maintaining Red data book.**

Ques6) Match the following:

1. MS Swaminathan – Green revolution
2. Methyl Isocyanate – Bhopal gas tragedy
3. Uranium- Nuclear energy fuel
4. Stratosphere- Ozone layer
5. Saprotrophs-Earthworms

2025 JANUARY

SECTION-A (ANY 3)

Q1.
a) Full forms (5*1=5)

b) True false (5*1=5)

Q2. Do any 4 (2.5*4=10)
Write short notes on the following:
1. Endemic species
2. Biological invasions
3. Global warming

4. Biodiversity
5. Swachh Bharat Abhiyan

Q3. Difference between (any 2) (5*2=10m)
1. Biosphere reserves and Biodiversity hotspots.
2. Resettlement and rehabilitation.
3. In-situ conservation and Ex-situ conservation

Q4. Explain the impacts of climate change on rural livelihood. Discuss the reasons why women in general are comparatively more impacted by climate change's impacts. (10m)

SECTION-B (any 2) (10*2=20)

Q5. Nuclear energy case study.

Q6. Population growth effects case study

Q7. What are the threats to biodiversity? Explain the various types of threats that facilitate habitat fragments, degradation and loss of natural areas. Also suggest some measures to prevent biodiversity loss.

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SECTION-A (ANY 3)

Q1. (10*1=10)
a) Full forms
b) Fill in the blanks

Q2. Do any 4 (2.5*4=10)
Write short notes on the following:
1. Endangered species
2. Carbon footprints
3. Montreal protocol
4. Sacred groves
5. Acid rain

Q3. Throw light on the major environmental movements witnessed in India. Also, discuss the economic and identity issues associated with environmental movements.

Q4. Do any 2
1. Discuss the impacts of climate change on agriculture for Indian context.
2. Explain the various level of biodiversity and their salient features.
3. Discuss the role of various cultural practices in environmental conservation.

SECTION-B (any 2) (10*2=20)

Q5. Hydropower energy generation case study, Carbon footprints.

Q6. Population growth case study

Q7. What do you understand by the value of biodiversity? Explain the various categories that summarise the value of biodiversity of terrestrial ecosystem (forest) ad aquatic ecosystem (riverine) with suitable examples.ouuu8

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